

INSTALLATION, OPERATION AND MAINTENANCE MANUAL

Generation III
Tri-Stack™ Fume Hood
Exhaust Systems

STROBIC AIR TECHNOLOGIES
A CECO Environmental Company

Rev. C 3/6/18

FACTORY TESTED

All fans are factory tested to ensure proper balance, alignment, air delivery, correct motor load, and correct working of all moving parts.

CUSTOMER INSPECTION

Equipment is designed to be removed from truck with a crane. A fork lift should not be used to pick up equipment from underneath units as they could damage internal components and/or paint. Strobic Air can, at a fee, properly skid equipment for fork lift removal, and/or crane yard storage, with written request prior to shipping.

PRIOR TO INSTALLING

- Inspect the equipment to determine if any damage has been incurred through mishandling during shipment. It is the customer's responsibility to notify the trucker immediately to claim for damages. Failure to properly notify trucker before signing off on reception of equipment will result in transfer of liability from trucking company to jobsite/parties responsible of equipment unloading.
- The name plate should be checked against the packing slip and job order for correctness. Some fans must be shipped in components due to trucker's height limitations. When this occurs and more than one fan of similar size are on the same project, carefully match and assemble components by the "line-up" tag (serial) numbers placed on each part.

Ensure the disconnect switch on the fans are bolted in place on the disconnect mount provided with the fan.

NOTE: It is not necessary to disconnect the wiring to the switch or motor in the terminal box, to do so ***will void the warranty.***





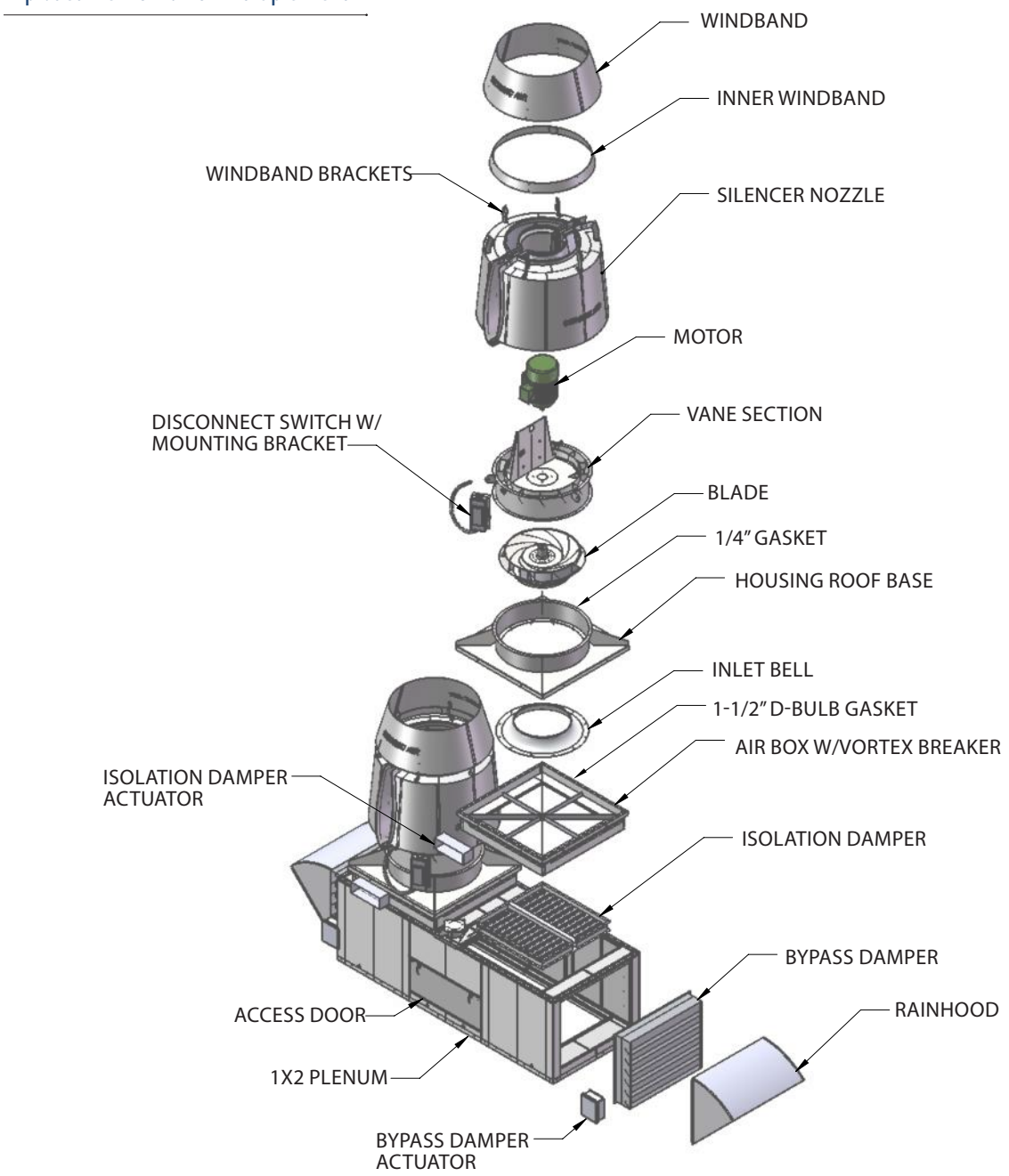


INSTALLATION INSTRUCTIONS

The TRI-STACK fan has been constructed in modules so it may be disassembled and installed in sections to minimize installation equipment requirements. Be sure all parts are aligned correctly per "line-up" tags and all gasketing is in place. (See customer inspection above). Each fan ships with required mounting hardware and tube of caulk taped to electrical conduit on fan. Each fan system has a box of touch up paint found inside the motor dish of one of the supplied fans.

Once the fan is in place, before the power has been connected, check the alignment of the blade and inlet bell. If power is already connected, be sure the power has been locked out. Remove the fan housing access door and spin the blade by hand. Check that no foreign materials such as loose nuts or bolts are inside the housing. Check that the blade spins freely and does not rub the inlet bell. If the blade rubs the inlet bell, contact your local representative for the Tri-Stack Field Blade Alignment Procedure. Once the blade and the housing have been checked, bolt the access door back on to the housing before starting the fan. Any unauthorized modifications to equipment without Strobic Air prior approval will void warranty.

Exploded View Of Fan On Multiple Plenum



TRI-STACK MOUNTING INSTRUCTIONS

CURBS

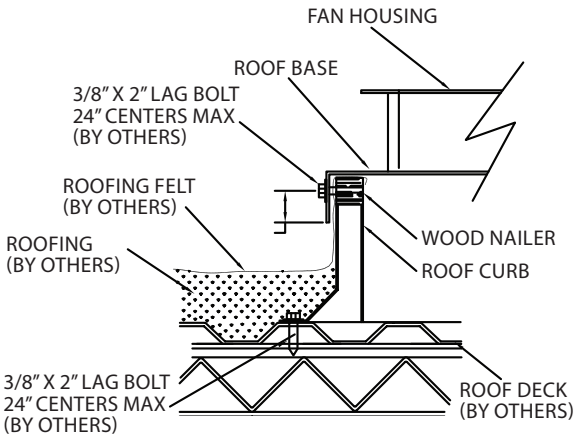
Curbs shall be a minimum of 14 gauge steel and anchored directly into the roof structure. The site structural engineer to determine site specific requirements for anchoring/supporting of equipment to building structure.

CURB CAP/PLENUM BASE

A square housing base with a 3" flange has been constructed for ease of installation to any frame opening. Refer to your certified drawing for other dimensions. Ensure that proper support is provided to support the weight of your fan. Ensure air leaks, if any, are properly sealed upon start-up. If the curb or structural base is not supplied by Strobic Air, consult the factory to avoid "soft base" vibration problems. Be sure the curb top or structural base is plumb and level to provide for proper drainage. The curb cap shall be appropriately lagged to the curb by means of 3/8" lag bolts or 1/4" bolts and nuts supplied by the contractor. (Heavier fasteners may be required by code in your area). A 316 stainless steel fastener is recommended for corrosion resistance. (See illustrations **C-1** to **C-9** for various base configurations).

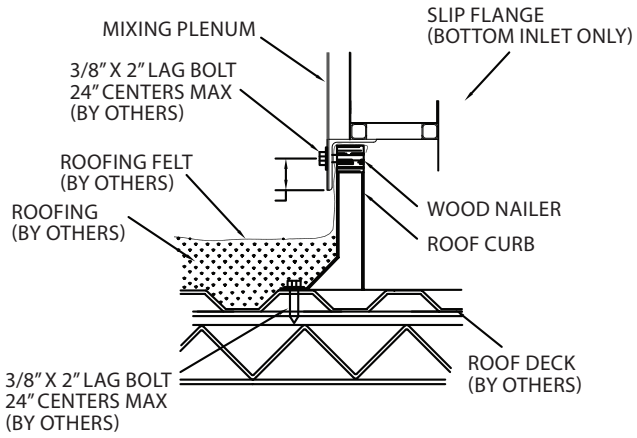
Single Fan Plenum to Roof Curb

C-1



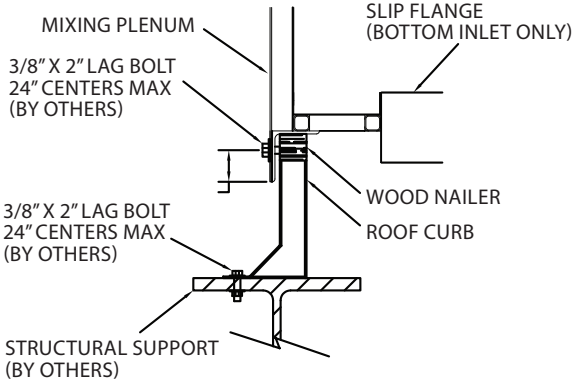
Multiple Fan Plenum to Roof Curb

C-2



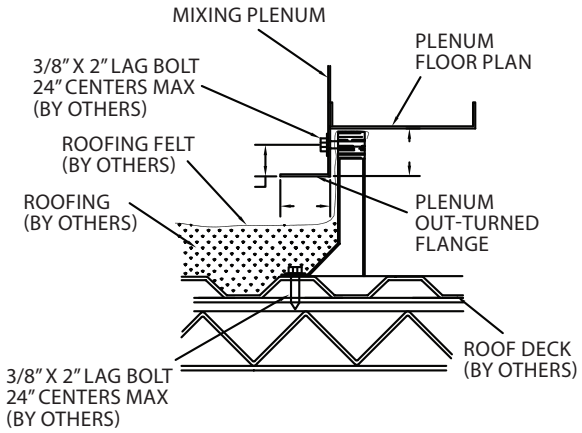
Roof Curb to Structural Steel

C-3



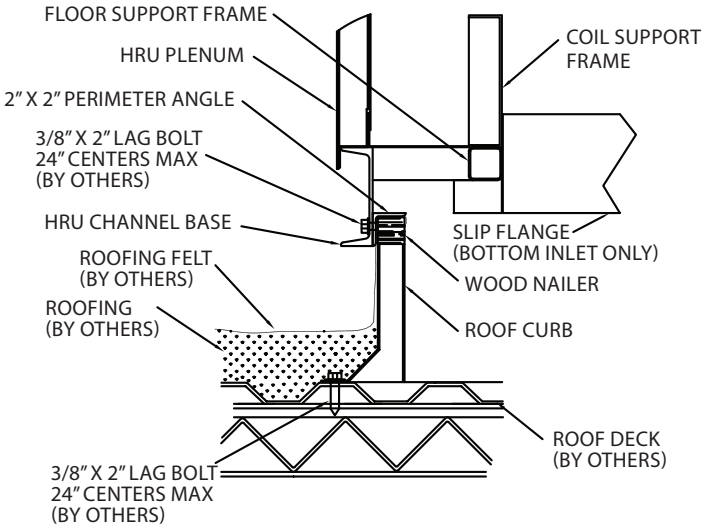
Multiple Fan, Single Wall Plenum to Roof Curb

C-4



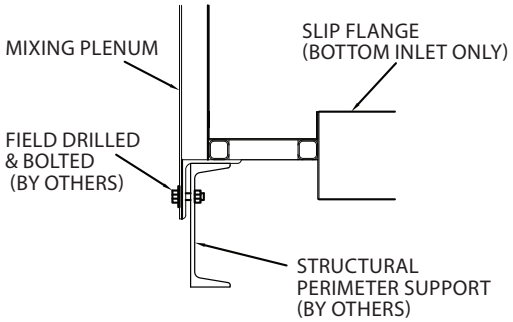
Horizontal HRU Plenum to Roof Curb

C-5



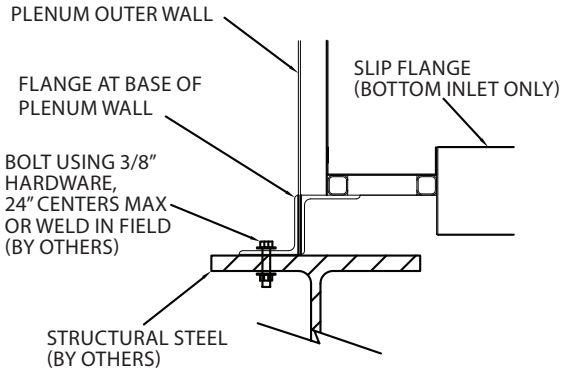
Plenum to C-Channel Base (or Concrete Curb)

C-6



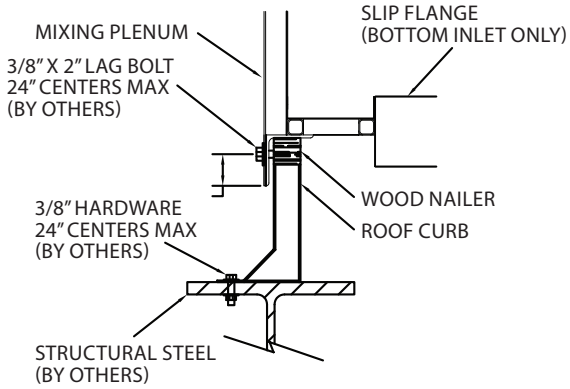
Double Wall Plenum to Structural Steel (or Concrete Pad)

C-7



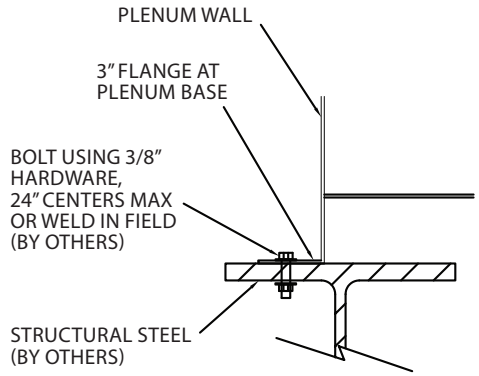
Roof Curb to Structural Steel

C-8



Single Wall Plenum to Structural Steel

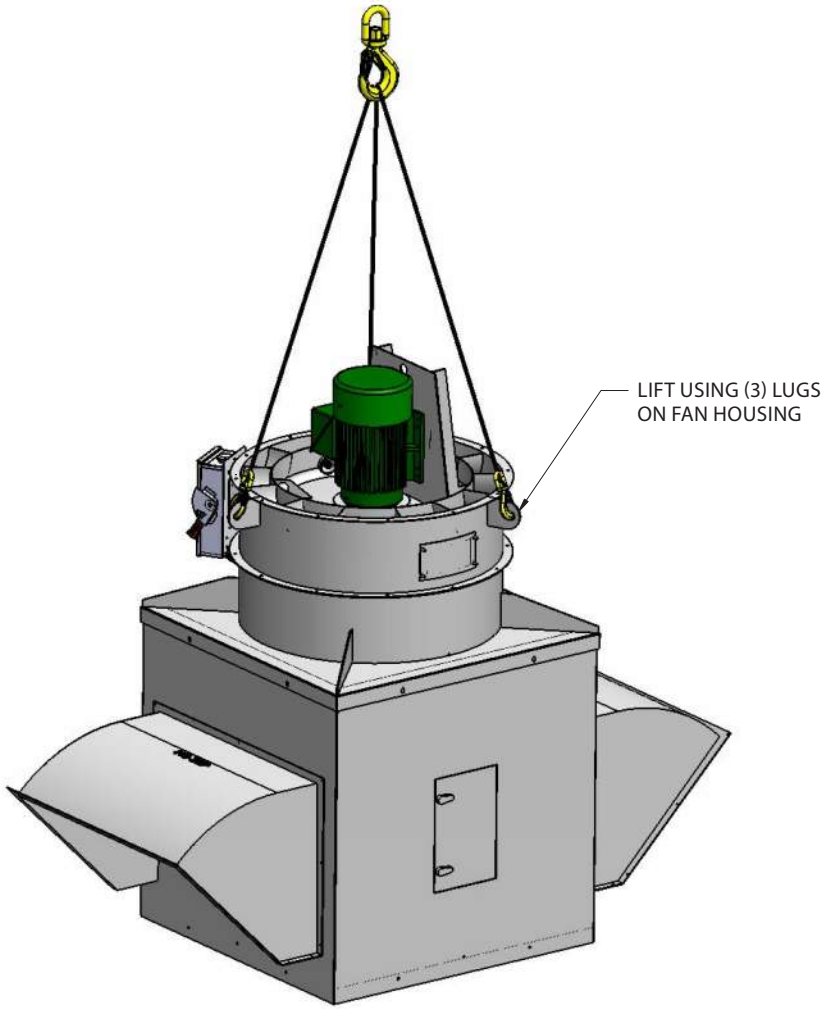
C-9



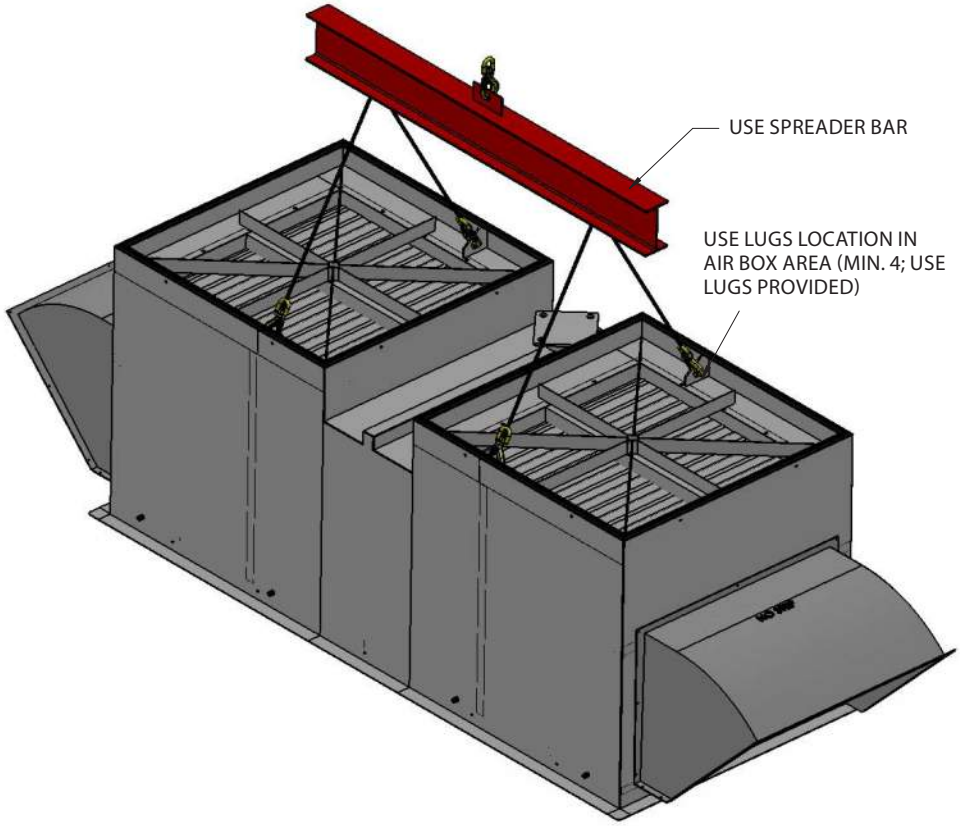
LIFTING INSTRUCTIONS

See Illustrations **D-1** through **D-8** for standard lifting methods of common Tri-Stack components.

Single Fan/Plenum Lifting Illustration D-1

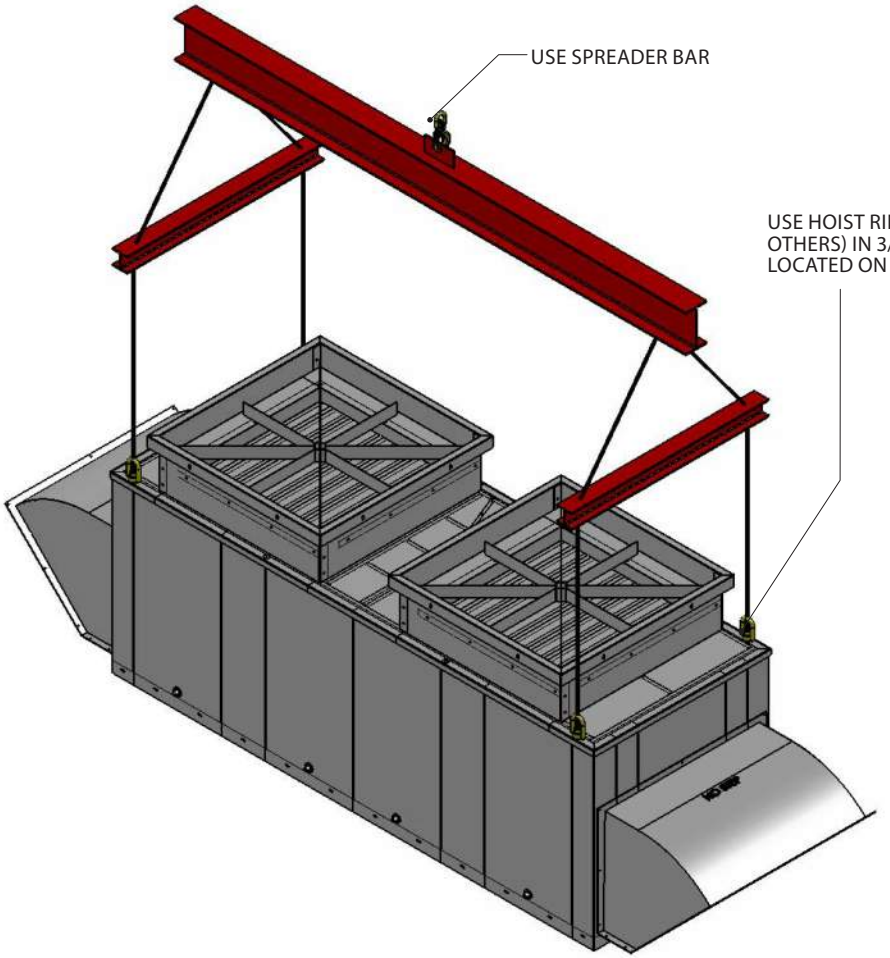


Single Wall Lifting Illustration D-2



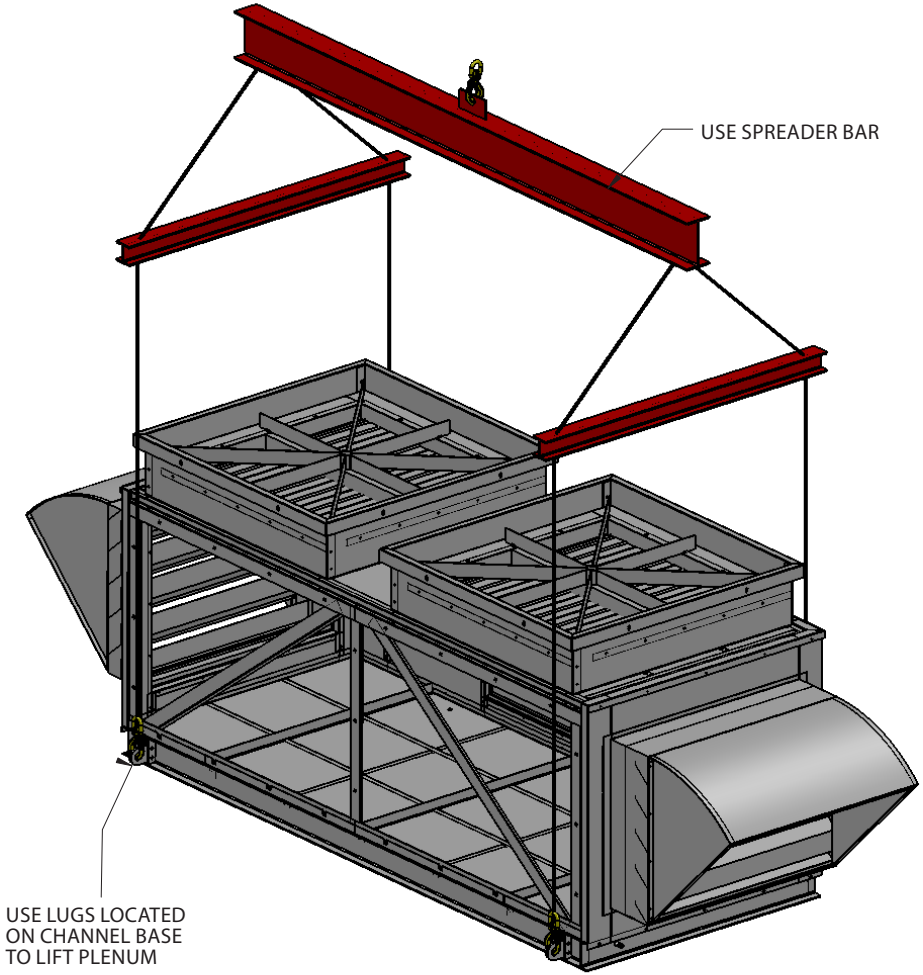
Double Wall Lifting Illustration

D-3



HRU Lifting Illustration (With Shipping Split)

D-4

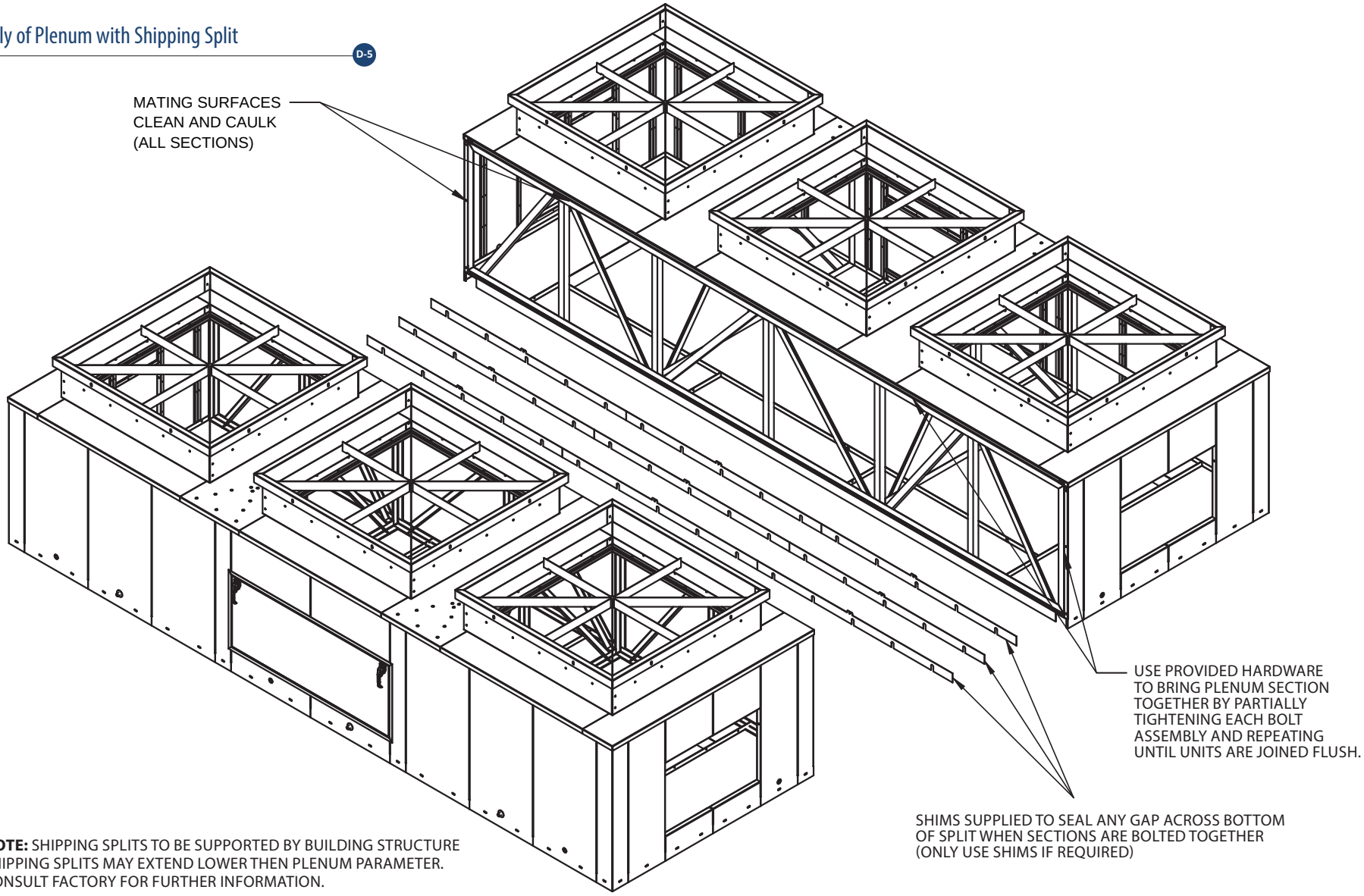


NOTE: SHIPPING SPLITS TO BE SUPPORTED BY BUILDING STRUCTURE.

LIFTING INSTRUCTIONS

Assembly of Plenum with Shipping Split

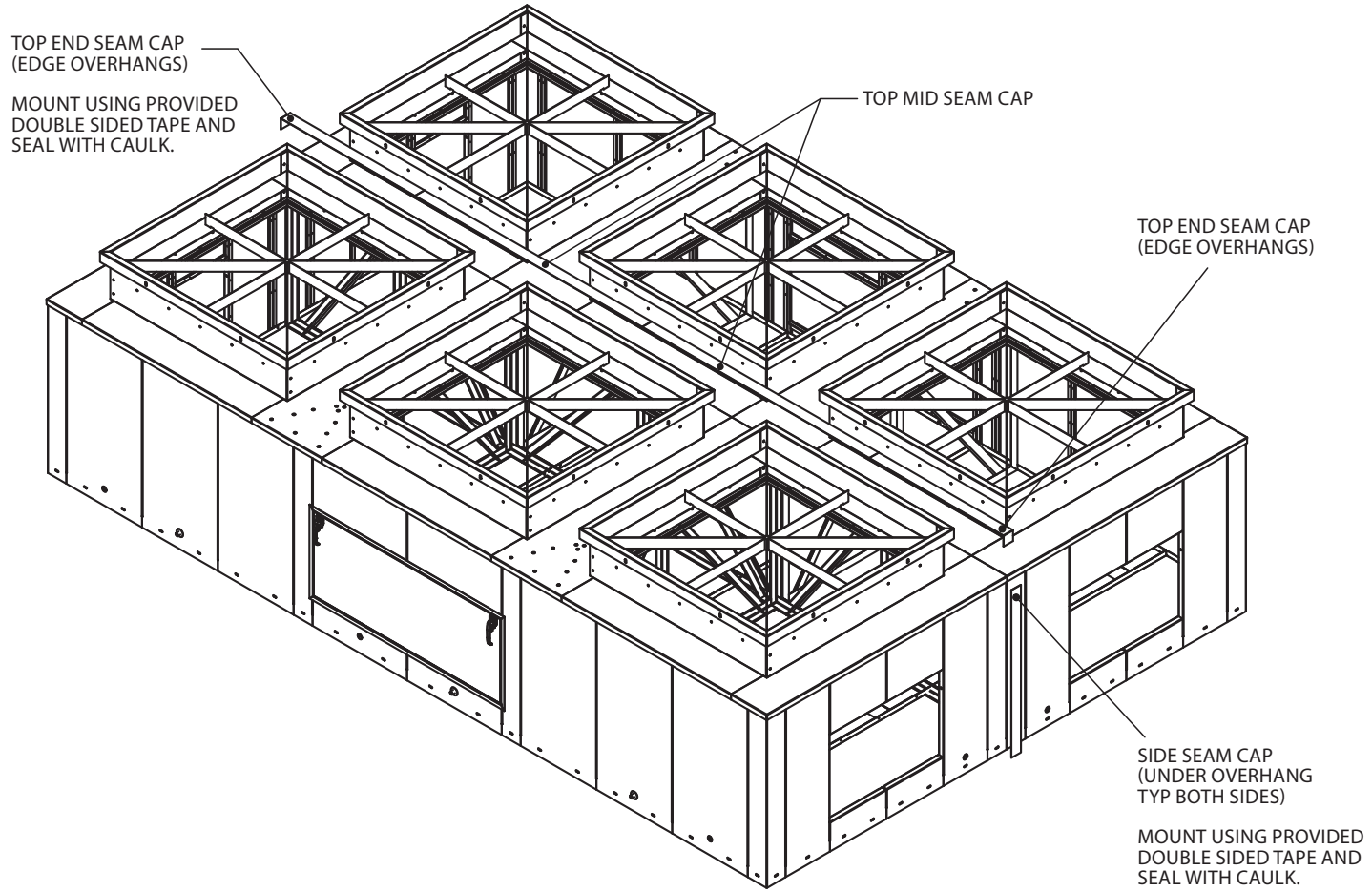
D-5



NOTE: SHIPPING SPLITS TO BE SUPPORTED BY BUILDING STRUCTURE
SHIPPING SPLITS MAY EXTEND LOWER THEN PLENUM PARAMETER.
CONSULT FACTORY FOR FURTHER INFORMATION.

Assembly of Plenum with Shipping Split

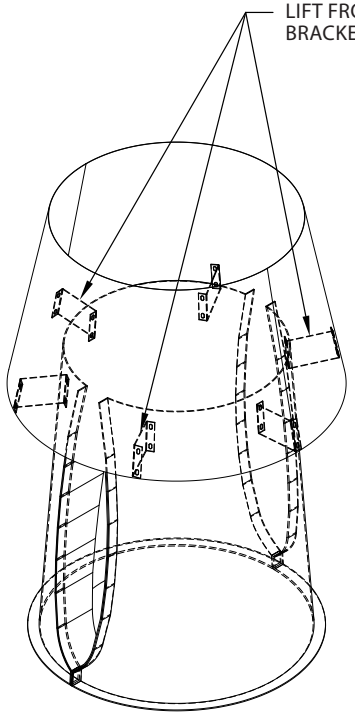
D-6



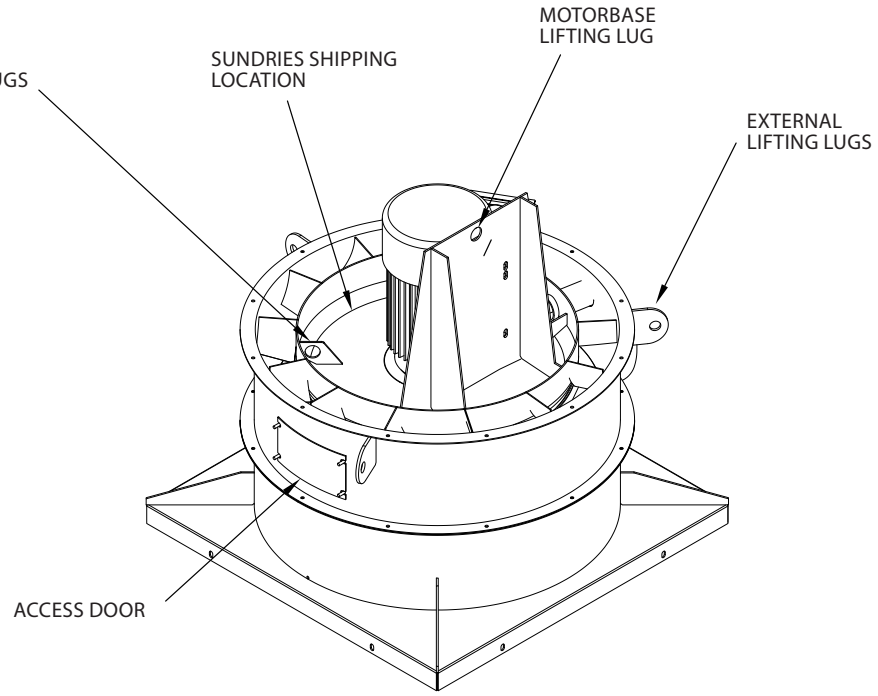
LIFTING INSTRUCTIONS

Lifting Points

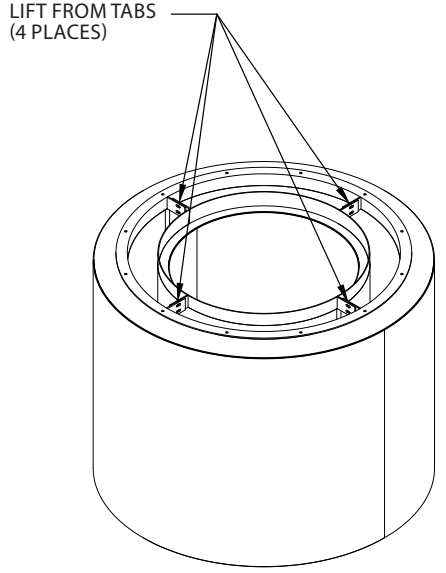
D-6



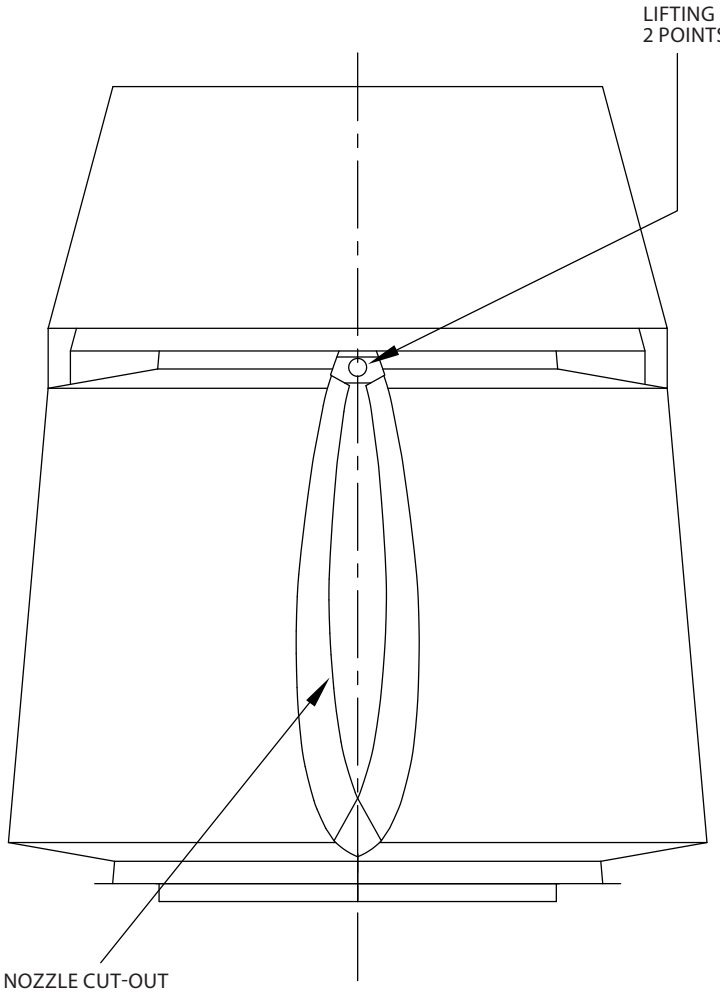
REGULAR NOZZLE



NOTES:
1. LIFT FROM EXTERNAL LIFTING LUGS (TYP-3)
2. IF NO EXTERNAL LIFTING LUGS, LIFT USING INTERNAL LIFTING LUGS (TYP-2) AND MOTORBASE LIFTING LUG
3. SUNDRIES SUCH AS TOUCH UP PAINT, HARDWARE AND CAULK SHIPPED INSIDE MOTOR BASE

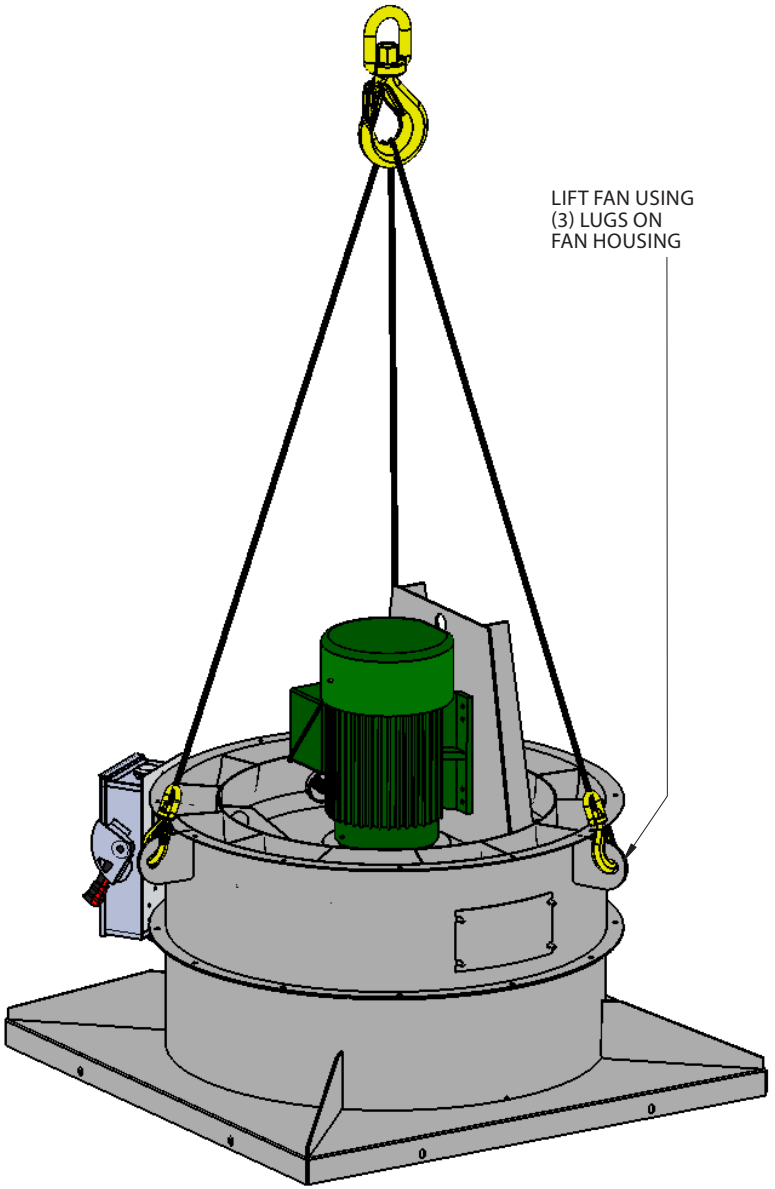


INLINE SILENCER



LIFTING INSTRUCTIONS:

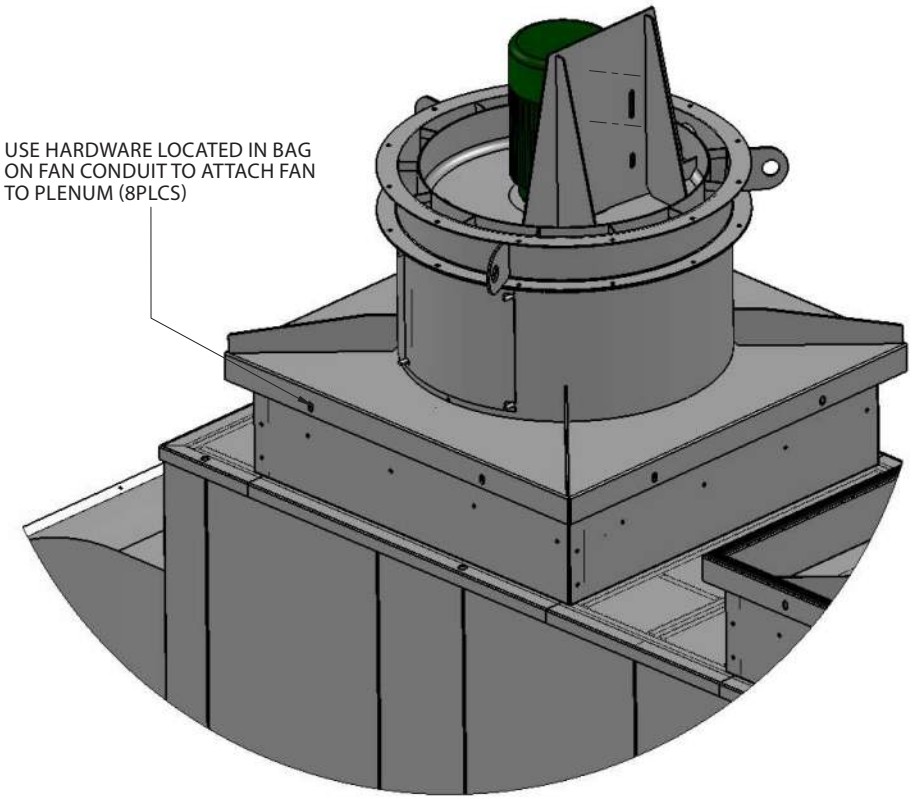
1. SILENCER TO REMAIN ON SKID UNTIL LIFTED ONTO FAN
2. USE SLING OR CHAIN TO LIFT SILENCER/NOZZLE FROM TWO LIFTING POINTS PROVIDED
3. SUNDRIES SUCH AS TOUCH UP PAINT, HARDWARE AND CAULK SHIPPED INSIDE MOTOR BASE
4. FOLLOW STANDARD CAULKING PROCEDURES FOR TRI-STACK NOZZLES



ASSEMBLY INSTRUCTIONS

Attaching Fan to Double Wall Plenum

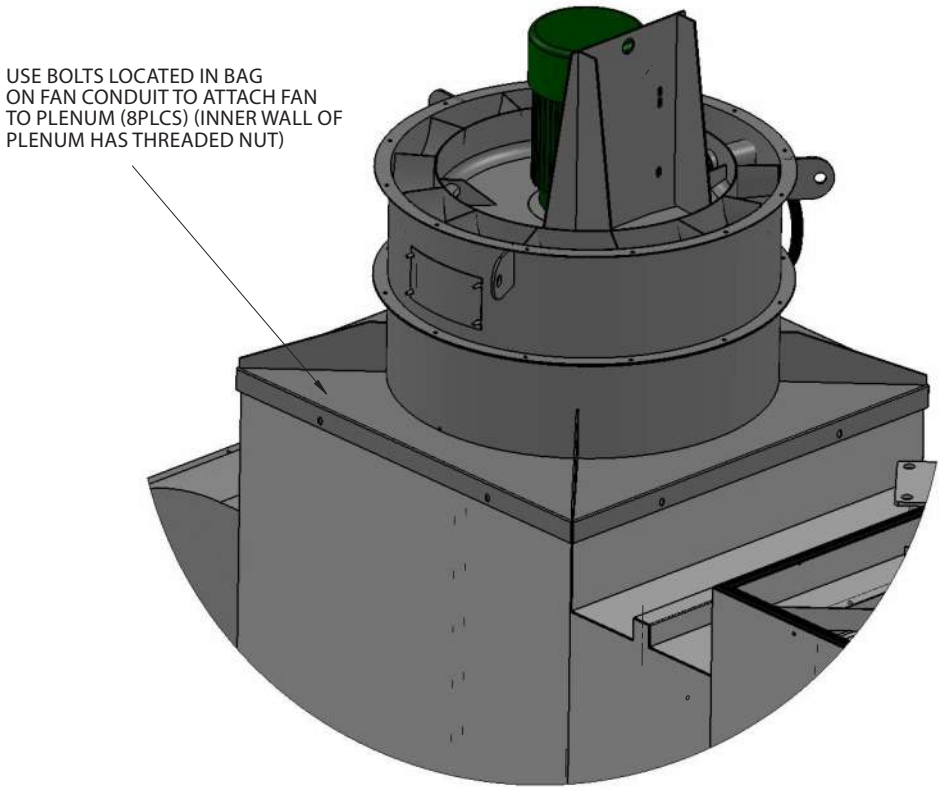
E-1



NOTE: USE LINE-UP MARKINGS ON PLENUM & FAN TO PROPERLY POSITION FAN ON PLENUM

Attaching Fan to Single Wall Plenum

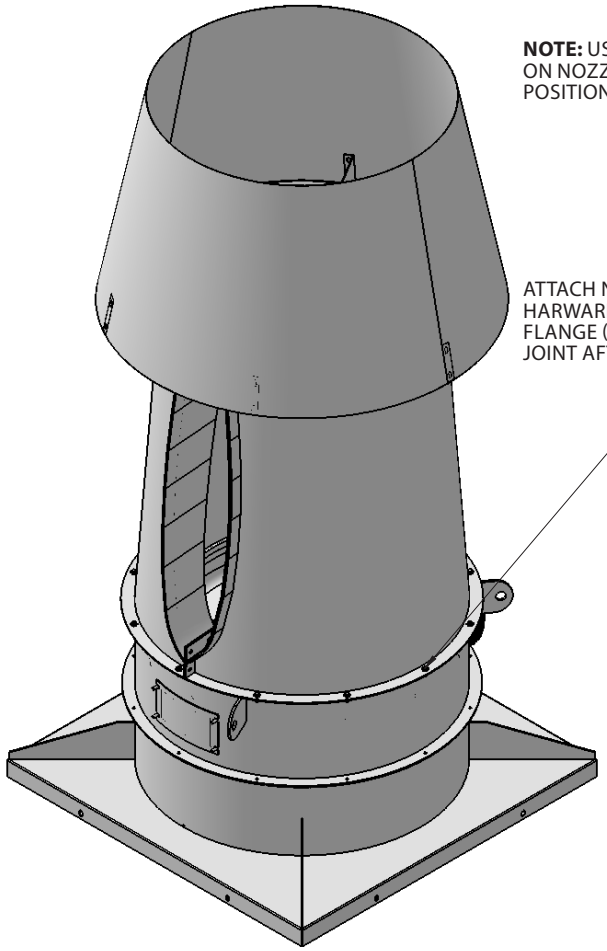
E-2



NOTE: USE LINE-UP MARKINGS ON PLENUM & FAN TO PROPERLY POSITION FAN ON PLENUM

Attaching Nozzle to Fan

E-3

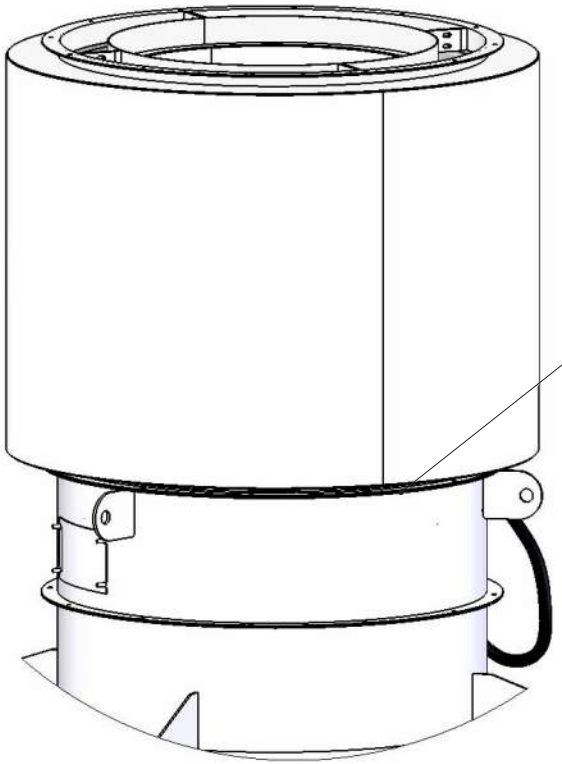


NOTE: USE LINE-UP MARKINGS ON NOZZLE & FAN TO PROPERLY POSITION NOZZLE ON FAN

ATTACH NOZZLE TO FAN USING HARWARE LOCATED IN FAN FLANGE (12 PLCS) (CAULK JOINT AFTER BOLTING)

Attaching Inline Silencer to Fan

E-4



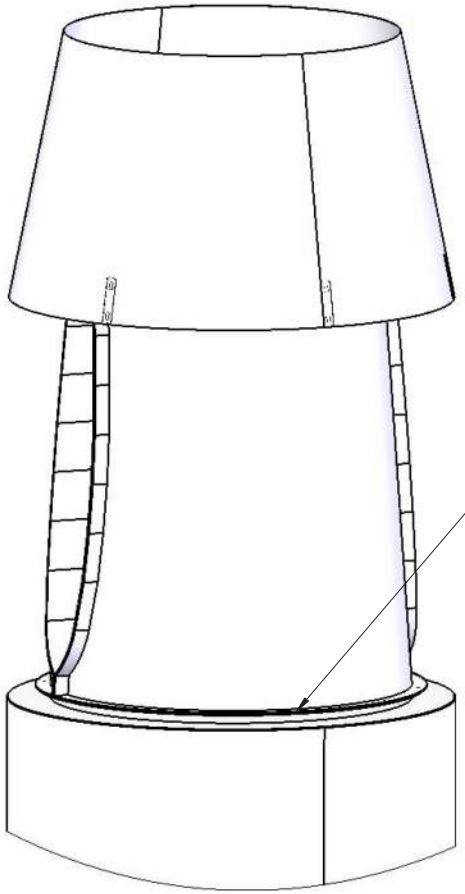
ATTACH SILENCER TO FAN USING HARWARE LOCATED IN FAN FLANGE (12 PLCS) (CAULK JOINT AFTER BOLTING)

NOTE: USE LINE-UP MARKINGS TO PROPERLY POSITION SILENCER ON FAN

ASSEMBLY INSTRUCTIONS

Attaching Nozzle to Inline Silencer

E-5



FIBERGLASS NOZZLE BOLTING TORQUE

Recommended torque values for stainless steel hardware when fastening a fiberglass nozzle to the fan flange are as follows:

BOLT SIZE	TORQUE
1/4 - 20	50 inch-lbs.
5/16 -18	75 inch-lbs.
3/8 -16	120 inch-lbs.

NOTE: Exceeding the given values by as little as 15% in some cases can cause minor cracking to occur around the nozzle flange and surrounding joints. Cracking and stress become more pronounced with further torque application.

Torque all hardware in steps to avoid damaging equipment. Start with one bolt and torque to 50% of full capacity. Work around unit until every bolt is at 50% total capacity. Repeat process at 75% and finally full torque rating.

STANDARD BOLT TORQUES

Recommended torque values for stainless steel hardware when fastening all-metal flanges are as follows:

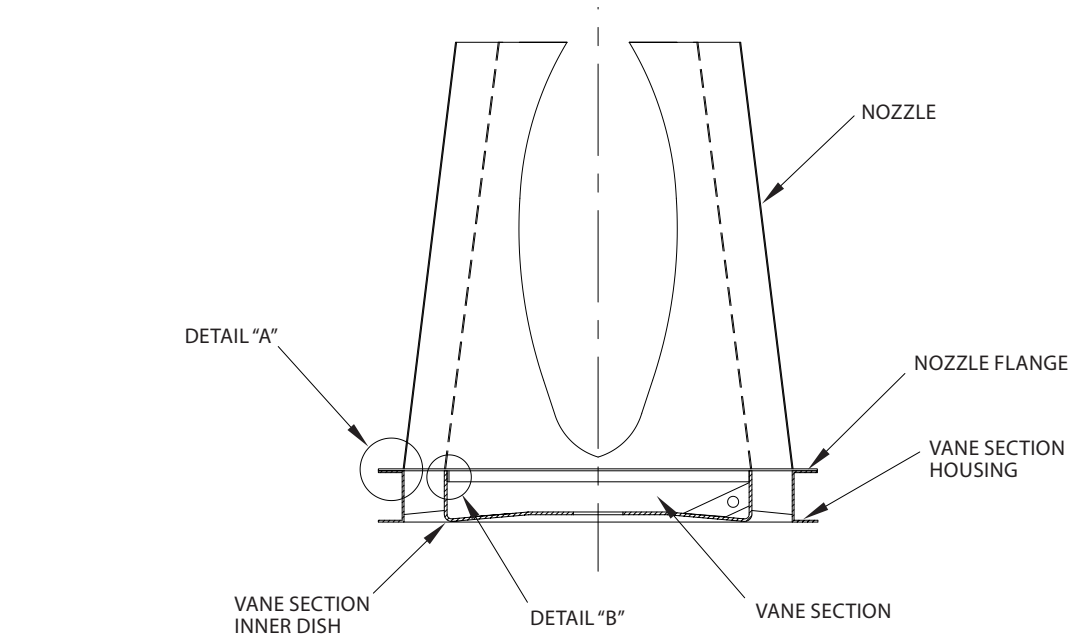
BOLT SIZE	TORQUE
1/4 - 20	79 inch-lbs.
5/16 -18	138 inch-lbs.
3/8 -16	247 inch-lbs.
1/2 - 13	542 inch-lbs.
5/8 -11	1160 inch-lbs.
3/4 -10	1582 inch-lbs.

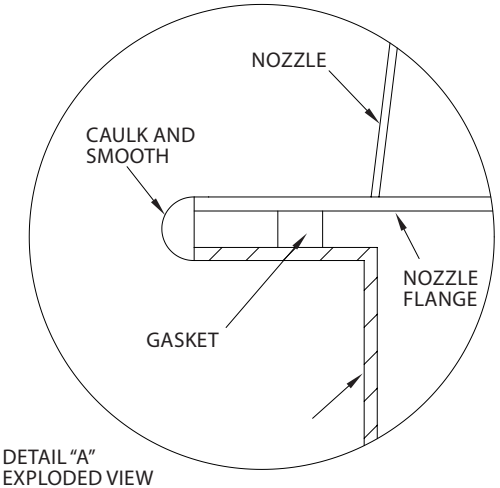
SEAL FLANGES

Fully caulk the outside edge of all the flanges with silicone caulk provided by Strobic.
(See Illustration F-1).

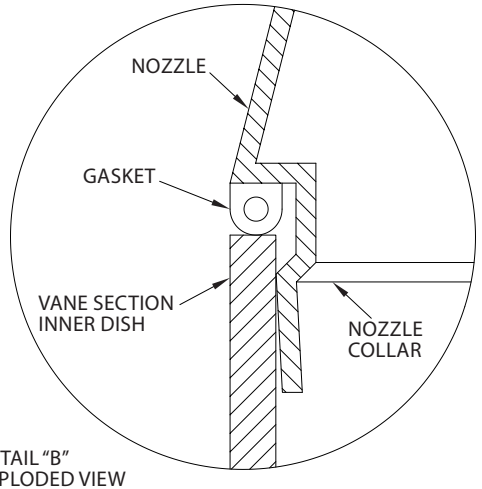
Nozzle to Vane Section Detail

F-1





DETAIL "A"
EXPLODED VIEW



DETAIL "B"
EXPLODED VIEW

MOTOR CONNECTIONS

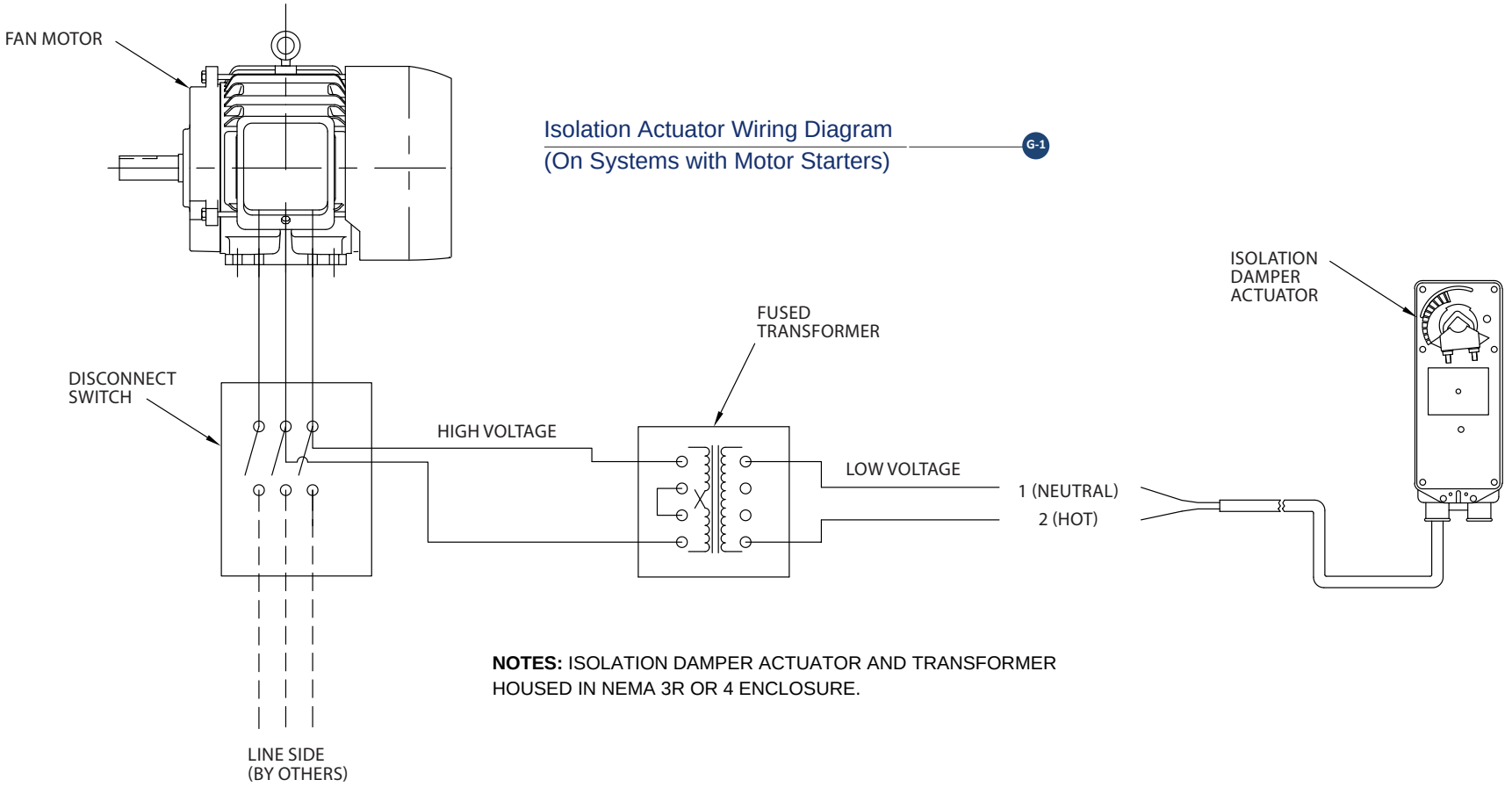
Unless specified differently, the motor on your fan has been wired to a NEMA 3R disconnect switch mounted on the side of the fan housing. Check the fan nameplate for proper voltage and amperage. If fans are stored for an extended period of time (180 days or more); all connections should be checked for looseness and corrosion.

ISOLATION AND BYPASS DAMPERS

Actuate the dampers slowly at start-up to observe they are free of binding and to ensure they open fully and close tightly. Adjust linkage accordingly.

ACTUATORS

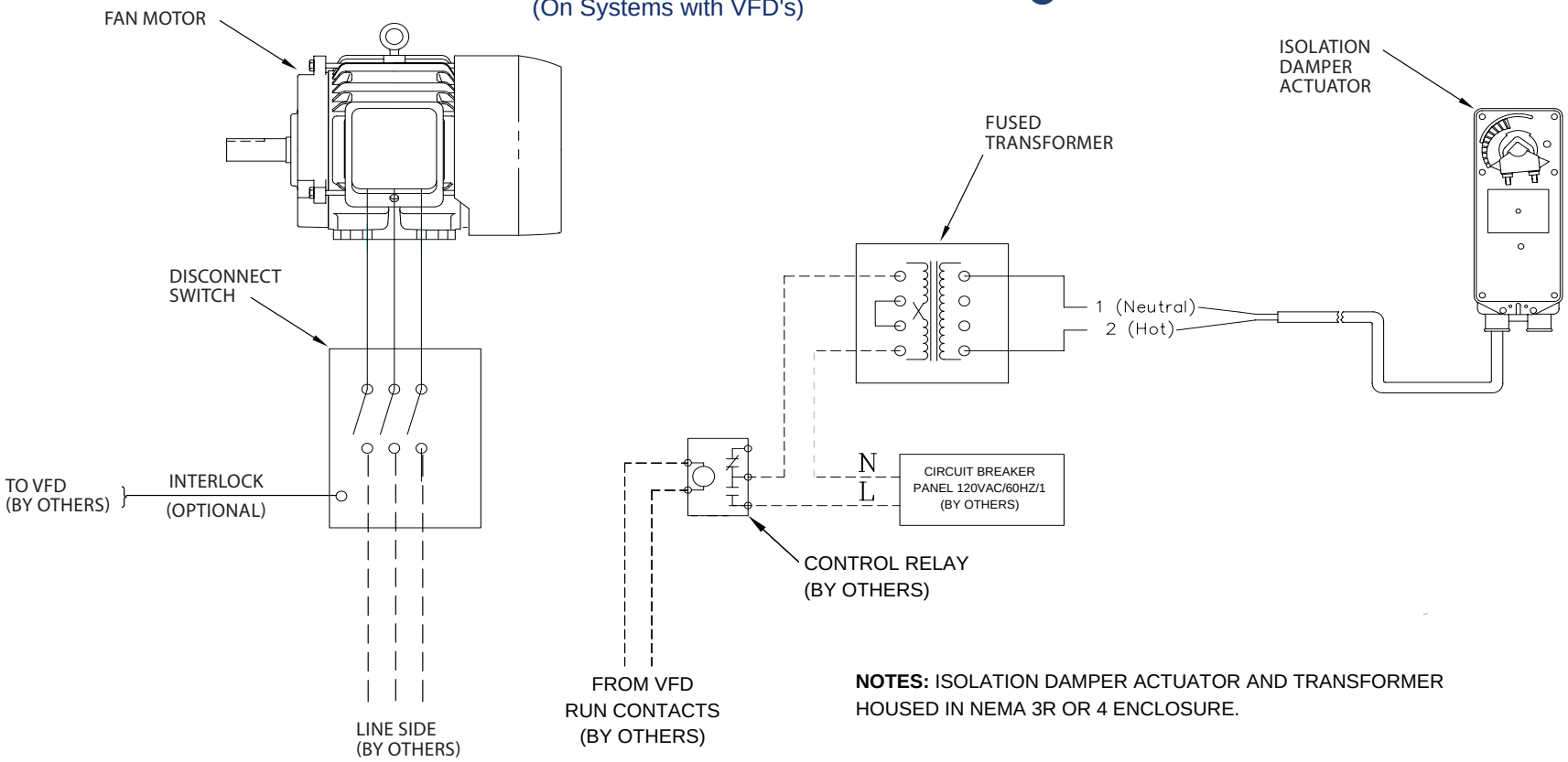
For pneumatic actuator, see separate instructions by manufacturer. Electrical schematics for typical actuators are shown on Illustrations **G-1 and G-2** (Isolation Actuator) and **G-3** (Bypass Actuator).



NOTES: ISOLATION DAMPER ACTUATOR AND TRANSFORMER HOUSED IN NEMA 3R OR 4 ENCLOSURE.

Isolation Actuator Wiring Diagram
(On Systems with VFD's)

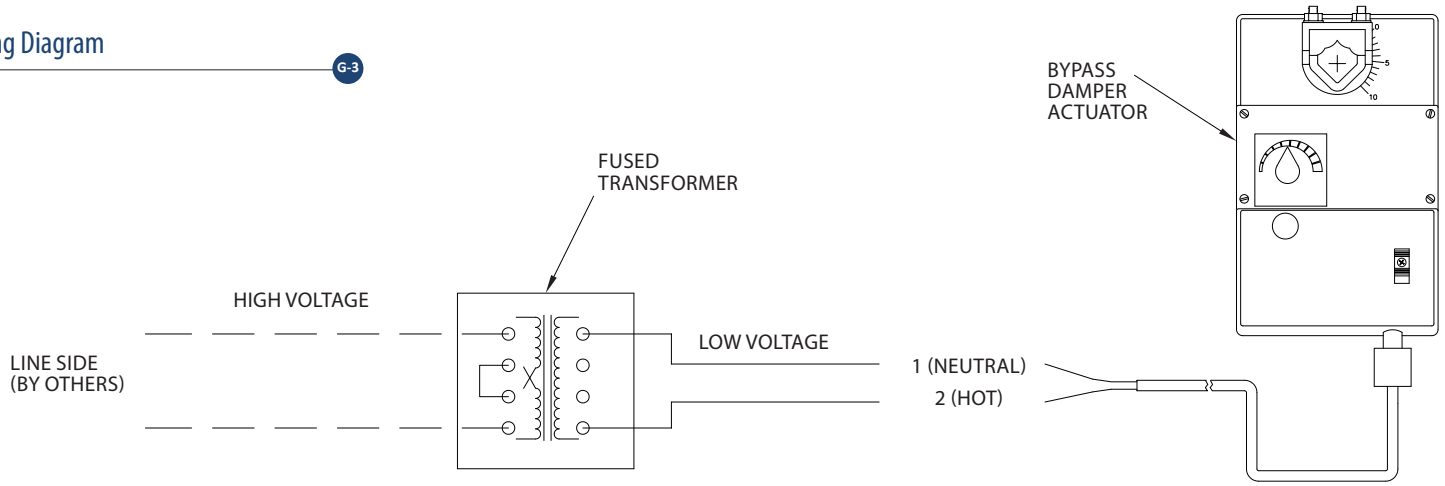
G-2



NOTES: ISOLATION DAMPER ACTUATOR AND TRANSFORMER HOUSED IN NEMA 3R OR 4 ENCLOSURE.

Bypass Actuator Wiring Diagram

G-3



NOTE: BYPASS DAMPER ACTUATOR RANSFORMER E ENCLOSED IN A NEMA 3R ENCLOSURE

EXTERNAL WIRING

This equipment must be installed with remote motor-overload protection. When connecting to a circuit protected by fuses, use time-delay fuses. Starting and overload control devices must be matched to motor rating. Follow the control manufacturer's instructions to make proper installation and connections. Connect electrical power supply to conform to National Electrical Code and any local regulations. Line voltage and wire capacity must match motor rating stamped on the nameplate.

VIBRATION ISOLATION

A 1-1/2" wide strip of neoprene bulb gasket has been factory installed on top of the mixing box. When the fan base must be shipped separately from the mixing box (due to transportation height requirements) care should be taken not to disturb this pad during installation. Check this point for air leaks upon start-up. Take necessary corrective measures to seal this joint properly.

INITIAL START

After installation is completed, but before fan is put in regular service, make an initial start as follows:

a. Hand turn the impeller to determine that the motor has not shifted in shipment and/or erection. The impeller should turn freely, should not touch other fan components and should be free of foreign materials.
b. Check that motor, starting and control device connections agree with wiring diagrams.
c. Check that voltage, phase, and frequency of line circuit (power supply) agree with fan nameplate.
d. Check that all bolts on your fan supports are tight, and that no objects are left in the fan housing.
e. Check motor service record and tag accompanying motor to be certain that bearings have been properly lubricated. When shipped from the factory, the bearings have been lubricated to give six (6) months satisfactory service except for sealed ball bearings which are lubricated for life and require no maintenance.
f. Momentarily energize the motor to check for proper rotation. The fan should be rotating clockwise from on overhead view, as indicated by arrow markings on the fan housing.
WARNING: Fans running in a reverse rotation can cause the fan motor to run into the service factor due to overamping thus the possibility of the circuit breaker tripping.
g. If motor is three-phase type, reverse rotation (if required) by inter-changing any two of the three power leads. If two-phase, inter-change stator leads of either phase, being careful not to change leads from one phase to the other. If motor is single-phase, refer to the instructions on the motor nameplate.
h. While the fan is running, check the mixing box, curb cap, inlet connections and fan/nozzle flanges for air flow leaks and make appropriate repairs/adjustments. Check with the factory if any leaks cannot be stopped.
i. Turn off fan and open access door on mixing box to observe proper full closure of isolation dampers. Make appropriate damper linkage adjustments. If adjustments are not easily implemented, consult the factory. If "fan staging" or a stand-by fan is employed, observe that the isolation damper under the "off" fan is fully closed while the other fans are in operation, and that no reverse "windmilling" is occurring. Make appropriate adjustments to the damper linkage.
j. Check the bypass dampers for proper operation. Be sure pilot positions are connected to the controls, when applicable.
k. Make appropriate linkage adjustments as outlined above.

WARNING: Repeated trial starts can overheat the motor (particularly for across the line starting) or increase the resistance of external starting equipment. If repeated trial starts are made, allow sufficient time between trials to permit heat to be dissipated from windings or external resistance and prevent overheating. Starting currents are several times running currents, and heating varies as the square of the current.

NOTE: Due to the utilization super premium efficiency motors the electrical circuit may see a large in-rush of current (approximately 19 times the FLA) for the first 1/2 to 1 cycle of the motor starting . For circuits that experience this condition we recommend the circuit breakers' **Instantons** Trip Setting be set to higher levels to mitigate against any nuisance trips.

VARIABLE FREQUENCY DRIVES

It is recommended that a start-up service technician from the drive manufacturer be employed. The full load amperage of the drive may differ from that of the motor. The lower limit of the electrical equipment should be observed.

As a rule of thumb, the motor lead lengths to the VFD should not exceed 200 feet on a 460 V system, and 100 feet at 575 volts. Noise can be amplified by lead length which can result in voltage spikes. If the motor is to be run above 60 Hz, close attention needs to be paid to the current in relation to the full load amperage (FLA) of the motor. Even though the motors can maintain constant horsepower up to 90 Hz, the torque capability of the motor will begin to drop off linearly to the increase of the speed. Therefore, if the torque drops off and the motor is expected to put out its rated horsepower, the current will start to increase linearly with the speed in order to meet the demand of the load. If the current increases higher than the rated FLA of the VFD, the VFD will begin to "trip-out" on an overload/over current condition. See specific drive manufacturer's installation, operation and maintenance manual for exact running condition details.

Motor shaft grounding kits or insulated motor bearings should be employed when using VFDs. Without such protection, static charges can build up and arc, which may cause motor bearing failure.

TRI-STACK FANS-SEQUENCE OF OPERATION

The following is a recommended sequence of operation for multiple Tri-Stack fans mounted on a common system.

System Start-up and Increasing System Demand:

- Before starting the system, ensure that all bypass dampers are fully open and all isolation dampers are fully closed. Failure to open the bypass dampers could result in excessive pressure in the system and may result in damage to ductwork or other system components. Failure to close all isolation dampers could result in excessive bypass into the system. This condition may also result in windmilling of non-energized fans which could result in damage to the fans.
- Energize the first fan. At the same time, send a signal to the isolation damper actuator to begin opening the damper.
- After the fan has achieved full speed and the isolation damper is fully open, begin to close the bypass dampers as required in order to increase the system pressure to the design set point.
- As the system demand increases, (an) additional fan(s) may be required. The fans should be started one at a time. As each additional fan is energized, the isolation damper to that fan should begin to open. Opening the isolation damper of a fan before it is energized could make system control unstable.

Decreasing System Demand:

- As the system demand decreases, the controls should open the bypass dampers to increase bypass flow.
- When the bypass dampers reach the full open position and more turndown is required in the system, de-energize 1 fan. At the same time, close the isolation damper for this fan. Allow time for the system to stabilize before starting or stopping additional fans.

In addition to the above comments, the sequence of the fans should be varied so that all fans operate approximately equal hours. It is recommended that the fans should be sequenced so that any fan does not remain idle for more than thirty consecutive days.

The above sequence is recommended for multiple fan systems utilizing bypass dampers. For fan systems using variable frequency drives, a similar approach can be taken. However, additional steps are required to ensure that all operating fans are running at the same speed and the outlet velocity of the system does not fall below safety/design requirements.

*See Appendix A for Start-Up Sheet.

SAFETY ALERT

This unit has been shipped in new and clean condition. However, the equipment is designed to handle fumes from a research laboratory hood or other unknown source. These fumes may be hazardous to your health. Therefore, we recommend that you:

DO NOT OPEN ANY ACCESS DOORS ON A FAN OR MIXING BOX without first consulting your Facilities Safety Officer. Special cleaning procedures may be necessary prior to working on the internal components of this unit.

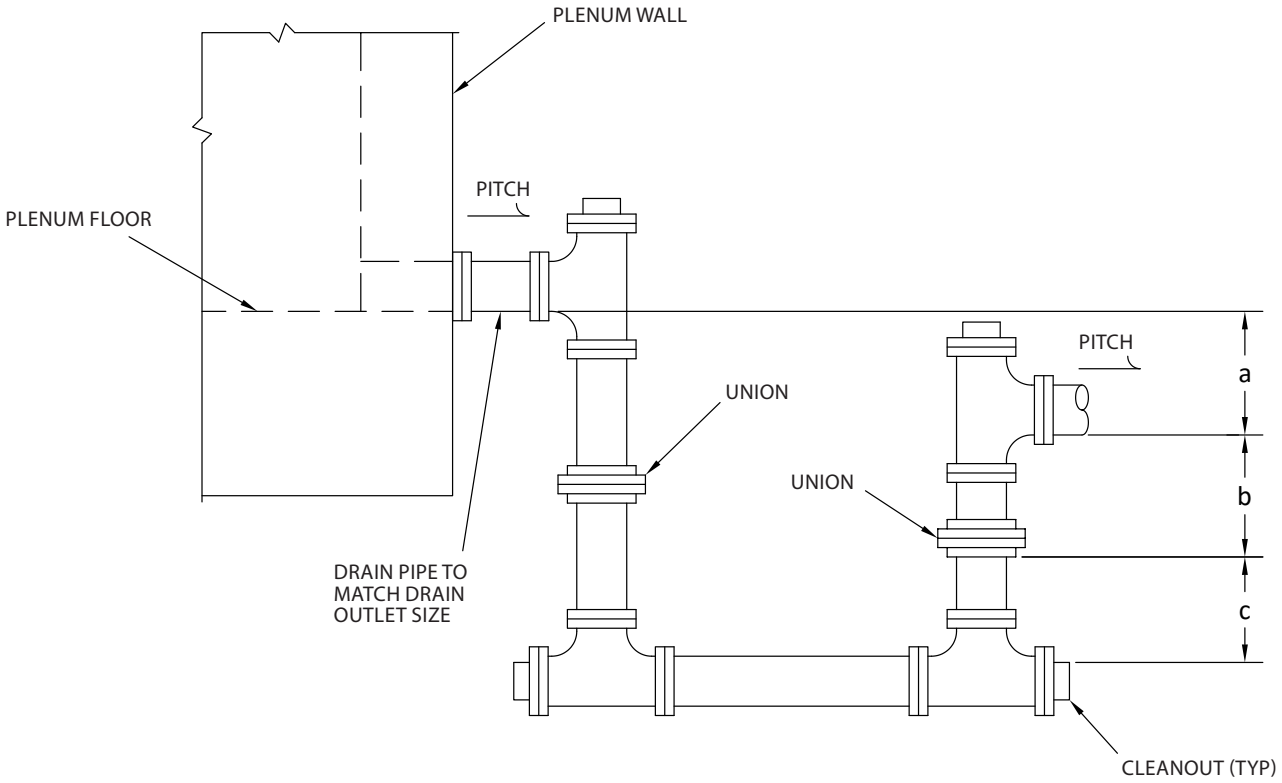
Without knowledge of the composition of the fumes exhausted through this unit, Strobic Air Corporation makes no general recommendation as to the cleaning procedure to be used. We are willing to consult with the Facilities Safety Officer and offer suggestions based on our field experience.

INTERNAL DRAINAGE SYSTEM

Weep holes are located in the motor cavity, in the fan housing near the inlet bell and in the mixing box/plenum. These holes should be checked periodically for possible clogging, especially when the fan is off. Should excessive pooling of water occur on the floor of the mixing box/plenum, it may be necessary to install an external drain (p-trap) on the weep holes. Illustration **H-1** details the recommended construction for a p-trap drain.

Recommended Tri-Stack P-Trap Drain Detail H-1

(Not Supplied by Strobic Air)



NOTES:
1. ALLOW SUFFICIENT SPACE BELOW THE PLENUM FLOOR FOR TRAP.
2. PITCH DRAIN FOR PROPER RUNOFF
3. SUPPORT LENGTHY DRAIN LINES TO PREVENT SAG AND RAIN WATER OVERFLOW
   (A) THIS DIMENSION IN INCHES MUST BE GREATER THAN THE MAXIMUM FAN SUCTION STATIC PRESSURE IN INCHES WC (WITH DIRTY FILTERS, COILS AND MAXIMUM AIR FLOW.)
   (B) THE DIMENSIONS IN INCHES MUST BE GREATER THAN 1/2 OF THE SAME MAXIMUM FAN.

REGULAR MOTOR MAINTENANCE

Maintenance of your STROBIC fan is limited primarily to the motor. Each motor should be inspected at regular intervals. The frequency and thoroughness will depend on the amount of operation, nature of service, and the environment. The motor exterior should be kept free of oil, dust, water, and chemicals. For fan-cooled motors, it is particularly important to keep the air intake opening free of foreign material. Do not block air outlet.

Motors will have the grease leads extended to the fan housing near the disconnect switch. For standard applications, the motor bearing should be lubricated with polyurea grease, such as Mobil Polyrex EM, as recommended by the motor manufacturer. For most general applications where the nature of service and environment is not severe, we recommend two (2) ounces of grease be supplied on each bearing in accordance with the following schedule:

SCHEDULE FOR LUBRICATION

900 RPM Fan Speeds	**Every 1-1/2 Years**
1200 RPM Fan Speeds	**Every 1-1/2 Years**
1800 RPM Fan Speeds	**Every 3 Years**

* May vary with motor manufacturer's requirements, consult motor manufacturer's instructions.

NOTE: The fan nameplate will show direct drive operating speeds and/or the actual motor speed. If there is difficulty with locating a proper acceptable equal to this grease or the specific grease mentioned above, please do not hesitate to consult the factory or your local agent. It is important to remember that **standard duty grease could cause shortened life to these bearings if used in lieu of the above mentioned grease or it's equal**. It is not necessary to open any access doors or remove any unit components to grease the motor bearings. **We recommend a grease gun be used, with a flexible nozzle, capable of being pressed onto a zerk type fitting.**

IMPELLER

Regular cleaning of the impeller is essential to maintain proper balance. The frequency of this cleaning depends upon the application. An improper balance condition due to accumulated dirt on your impeller can cause an overloaded condition in your motor or unnecessary vibration resulting in loose mountings, bad bearings, or broken blades and housing.

STORAGE OR DOWN TIME

If your fan is to sit idle for more than 30 consecutive days during start-up or shut-down, **especially in extreme cold and icy weather** care should be taken to hand spin to prevent "Peening" or "Brinelling" of the bearings. If the fan is outside, exposed to the elements, care should be taken to step up this procedure. Under no circumstances should a fan be exposed to a hostile environment without adequate protection. Common sense should prevail. When fans are to be placed in service after an extended shutdown or delayed start-up, motors should be meggered every 6 months in accordance with the motor manufacturer's start-up procedure. A minimum of 10 megaohms are recommended. In addition, all electrical connections should be checked for looseness and corrosion. The procedure for initial start-up should then be followed.

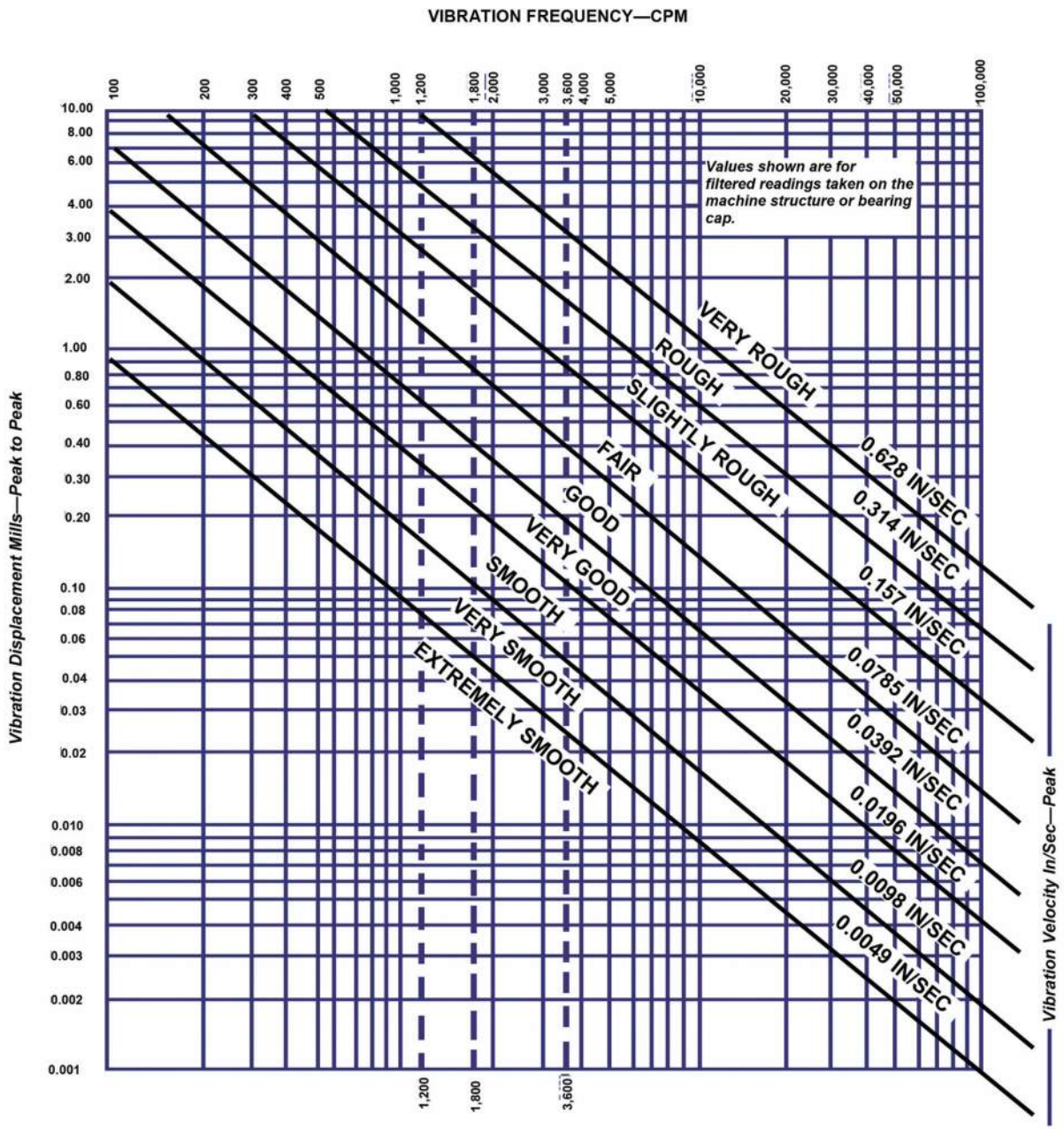
VIBRATION MONITORING

We recommend that the fan be monitored for vibration on a periodic or continual basis. Periodic checks by portable vibration equipment are recommended for non-critical installations with minimal system effects. The lubrication schedule could serve as a timely vibration monitoring schedule. A significant change in vibration levels will serve as notice for scheduling preventive maintenance.

CONTINUOUS VIBRATION MONITORING can best be accomplished by installing a vibration switch on the side of the fan housing. Inexpensive switches can provide both a warning system (remote indicator light) and a trip-out system in case of bearing failure. This system is recommended for critical installations or where adverse system effects are present. For commercial installations, we recommend a warning setting of "slightly rough" for a period of twenty (20) seconds, and a trip-out setting of "rough" for a period of five (5) seconds, as shown on the "General Machinery Vibration Severity Chart", Figure **I-1**.

General Machinery Vibration Severity Chart

I-1



PROCEDURE FOR MOTOR CHANGE OUT ON TRI-STACK FANS

The following are the standard instructions as recommended by Strobic Air Corporation for the rare event when a motor change of a TRI-STACK fan is required. These guidelines utilize standard lifting, rigging, and assembly procedures.

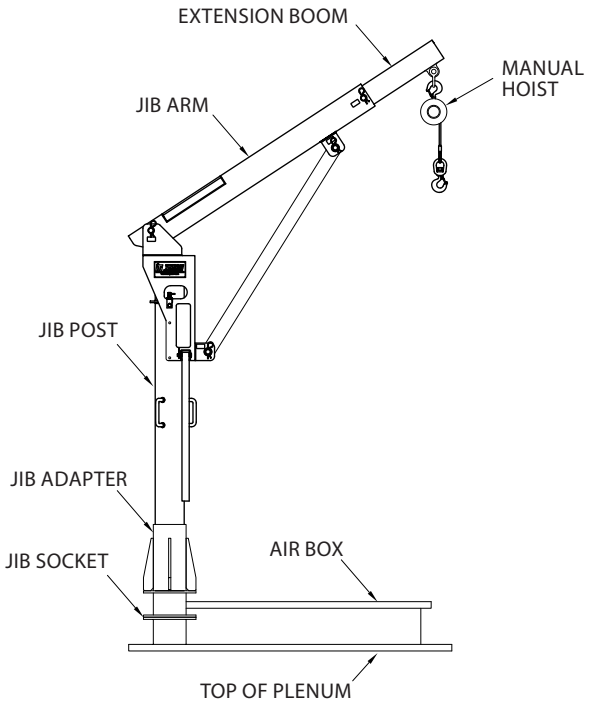
1) Isolate the electrical power supply with use of the disconnect switch and lockout device. Disconnect the electric supply between the fan motor and disconnect switch.
2) Remove the stainless steel nuts, bolts, flat, and lock washers holding the fiberglass stack to the vane section (3/8"-16 x 1-1/2"). Mark mating flanges for proper fit up upon assembly.
3) Remove the fiberglass stack leaving the windband attached to the stack.
4) Remove the stainless steel nuts, bolts, flat, and lock washers holding the vane section to the blade housing (3/8"-16 x 1-1/4").
5) Remove the motor, vane section, and fan blade as one unit.
6) Stand fan blade assembly in a horizontal position with the motor base resting on the floor. Block the unit with wood under the motor base and outside the vane section to secure the unit, and keep it from rolling.
7) Remove the keeper bolt's locking tab and plate that attach the fan blade to the motor shaft (3/4"-10 x 3").
8) Remove the impeller and motor key. A hub/blade puller might be required to remove the impeller.
9) Remove the four c-face bolts (5/8"-11 x 2").
10) Remove the four motor foot bolts (3/4"-10 x 2-1/2").
11) Remove motor.
12) Install new motor by securing the four c-face bolts. Place bolts in foot mount, but do not tighten.
13) Re-install the impeller, motor shaft key, keeper plate and associated hardware.
14) Scrape old gasketing from mating surfaces and lift vane section assembly onto the blade housing with new gasketing pre-installed on blade housing mating flange.
15) Align holes to maintain the correct orientation of the motor junction box. Place hardware into holes but do not tighten. Check that the blade spins freely without interfering with the inlet bell. After positive confirmation of clear blade path, tighten all hardware.
16) Install new gasketing on vane section and stack mating surfaces. Maintain correct orientation between the cutout of the fiberglass nozzle and the junction box of the motor. Tighten associated hardware.
17) With use of feeler gauges, check for soft bases between the motor foot and the motor base.
18) Shim accordingly with paper shims to maintain a 0.002-0.005 tolerance. Tighten the four foot mount bolts.
19) Using silicone caulk, seal all mating surfaces that have been separated including the connection of inner vane section and the fiberglass stack.
20) Re-connect the electrical wiring and conduit from the motor box and disconnect switch.
21) Unlock the isolated disconnect switch and "bump" the motor to verify proper fan blade and motor rotation.
22) If rotation is correct start the fan. When the unit is up to speed, check amp, voltage, and vibration readings. Consult the factory if unit is not performing to specification.

PROCEDURE FOR TRI-STACK MOTOR CHANGE OUT USING BOLT-ON JIB CRANE

1) Disconnect all electric to the fan disconnect switch using standard lock out procedures.
2) Unbolt the disconnect switch from the vane section bracket and place it by the motor to allow for the nozzle to be removed without any obstruction.
3) Bolt jib post to the jib socket with 1/2 - 13x2" bolts. If the jib has a post instead of a plate, slide the post through the socket. (J-1)
4) Unbolt the nozzle from the vane section. Rotate the nozzle in place to break any adhesion to the vane section caused by caulk or gasket. Wrap the hook from the jib arm around one of the windband brackets. (J-3)
5) Lift the nozzle away from the fan using the lifting arm and lower it to the side of the plenum. (J-4)
6) Use the two lugs near the motor and the hole at the top of the motor base as the three picking points.
7) The fan must be raised level, be sure that the chain at the hole in the motor base is short enough to provide a level and even lift. (J-3)
8) Unbolt the housing from the vortex breaker. (J-3)
9) Slowly raise the fan with the hoist. The fan only needs to be raised 2". Rotate the fan away from the plenum and lower it. (J-5)
10) Lower the fan onto plywood or a wooden pallet. (J-6)
11) When the fan has been fully lowered, unbolt the housing flange to the vane section. (J-6)
12) Raise the fan once more, pull the housing away and lower the fan back down. The fan is now resting on the inlet of the fan blade. (J-6)
13) Remove the chain from the lifting point on the motor base leaving two chains attached to the two lugs by the motor.
14) The fan needs to be tipped backwards and lean against the motor base. To do this, slowly lift the fan from the two lifting points. This can cause the fan to swing, be careful that the fan is clear of any objects.
15) Lower the fan and rest it on the motor base and the vane section top flange. The bottom of the motor should be facing the roof. (J-7)
16) The blade is secured to the motor shaft by a single bolt keeper plate. Remove this bolt and pull the blade. (The jib can be used to assist this step if needed.)
17) Secure the jib hook to the eye-bolt of the motor. Slowly rotate the jib crank just until the cable is tight.
18) Unbolt and remove the motor.
19) Bolt the new motor in place but do not fully tighten the bolts for the motor feet.
20) Use feeler gages to check the clearance between the motor base and the motor feet. Use paper shims to maintain a 0.002-0.005 tolerance. After shimming the motor base, tighten the bolts for the motor feet.
21) Line up the keyway in the shaft with the keyway in the blade hub and install the blade. Be positive that the shaft key is in place prior to bolting on the keeper plate.
22) Pick up the fan again using the two lifting points. Now rotate the fan and lower it so it rests on the inlet of the fan blade again.
23) Place new gasket on the vane section flange, vortex breaker flange and housing flange.
24) Now lift the fan again from the three pick points and install the housing. Be sure that the inlet bell is centered in the blade inlet and that the blade can spin freely before tightening the bolts on the flange. (The alignment of the bell and blade can be done with the use of drift pins in the flange holes.)
25) Any wiring that needs to be done to either the motor or the disconnect switch should be done now before the nozzle is in place.
26) Place the fan back onto the vortex breaker and bolt into place.
27) Bolt the nozzle and the disconnect switch back in place.
28) Using silicone caulk, seal all mating flanges.
29) Run wires to the fan disconnect switch and "bump" start the fan to check the blade rotation is correct.
30) Once the rotation is correct, and the fan is up to full speed, check amps, volts and the vibration readings. Consult the factory if the fan is not performing to specification.

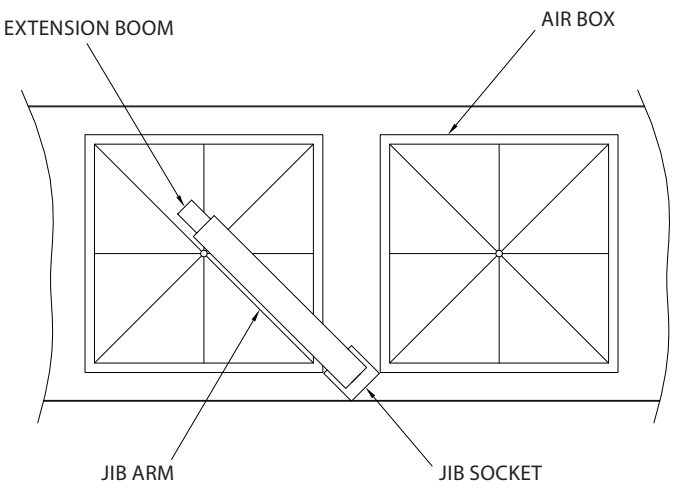
Side View of Jib on Plenum

J-1

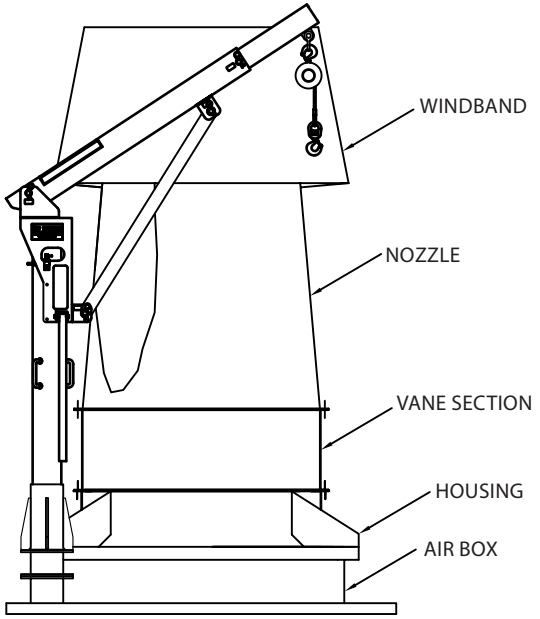


Top View of Jib on Plenum

J-2

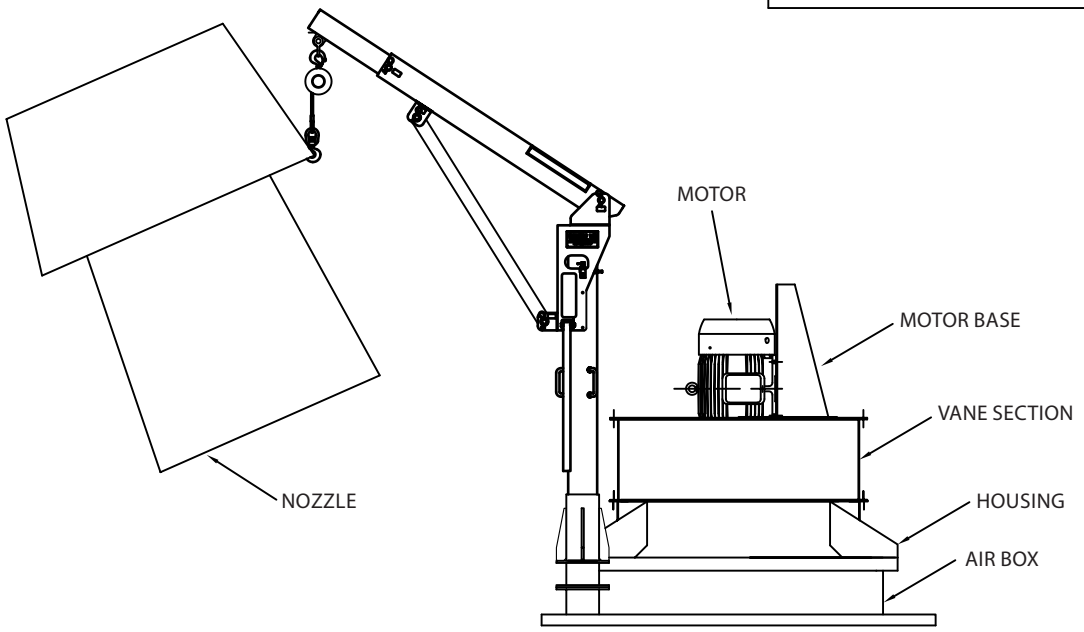


J-3



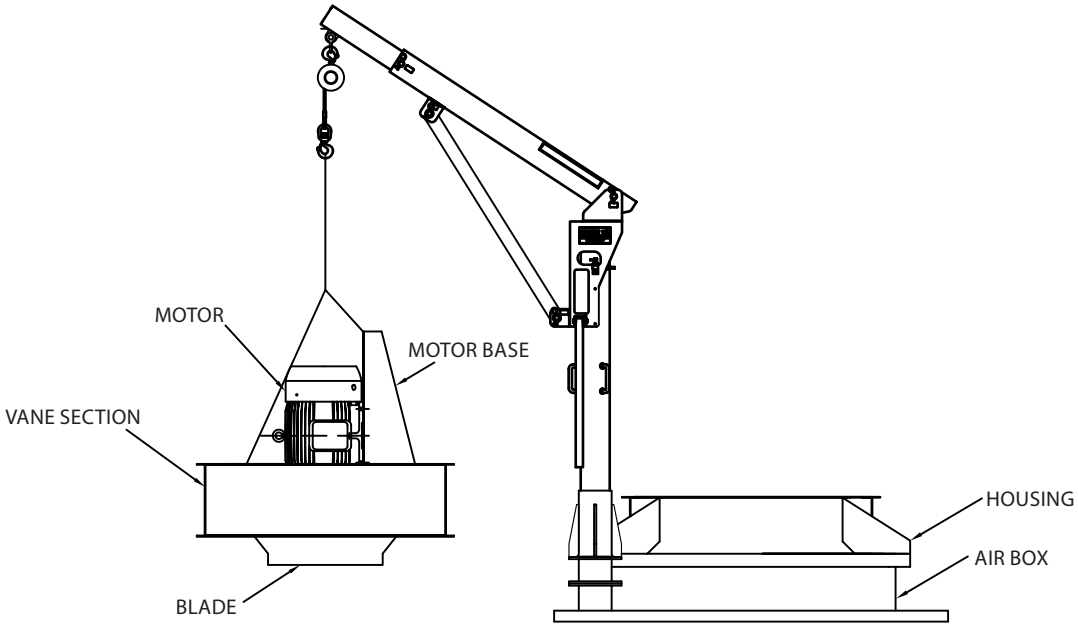
Lift Nozzle High Enough to Clear Motor Base & Motor

J-4



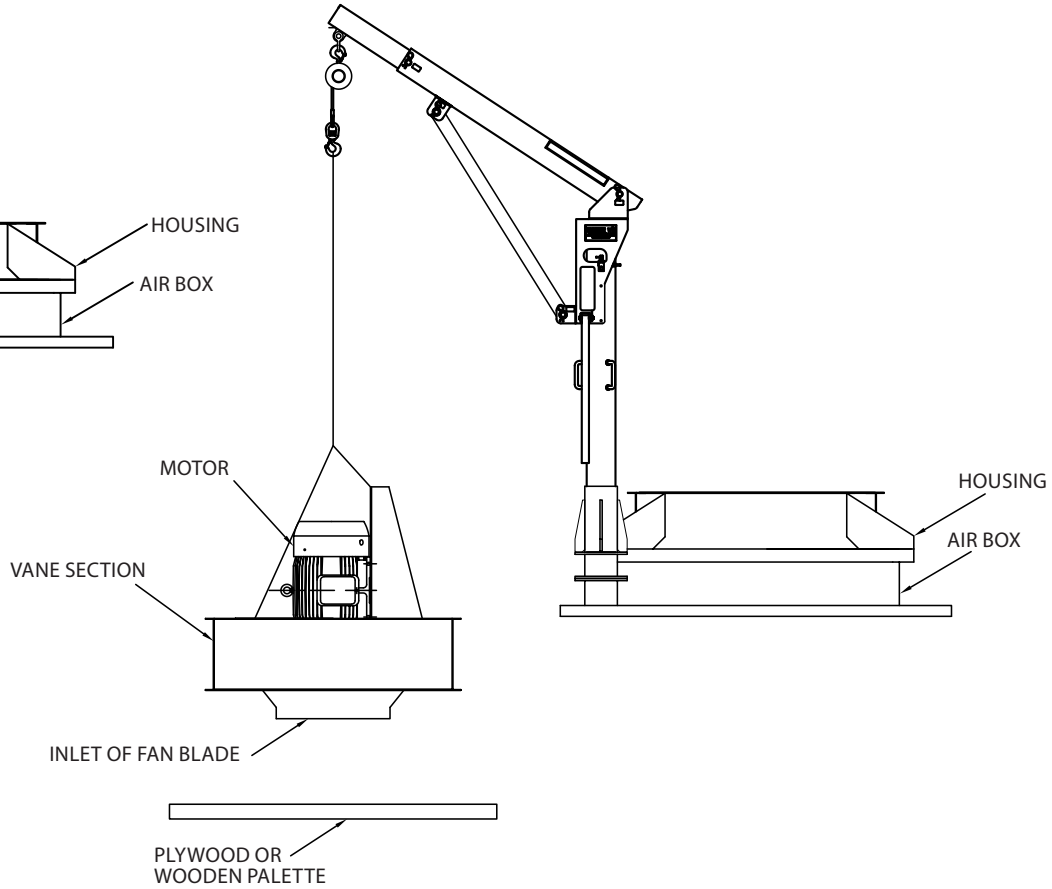
Lift Assembly High Enough to Clear Fan Housing

J-5



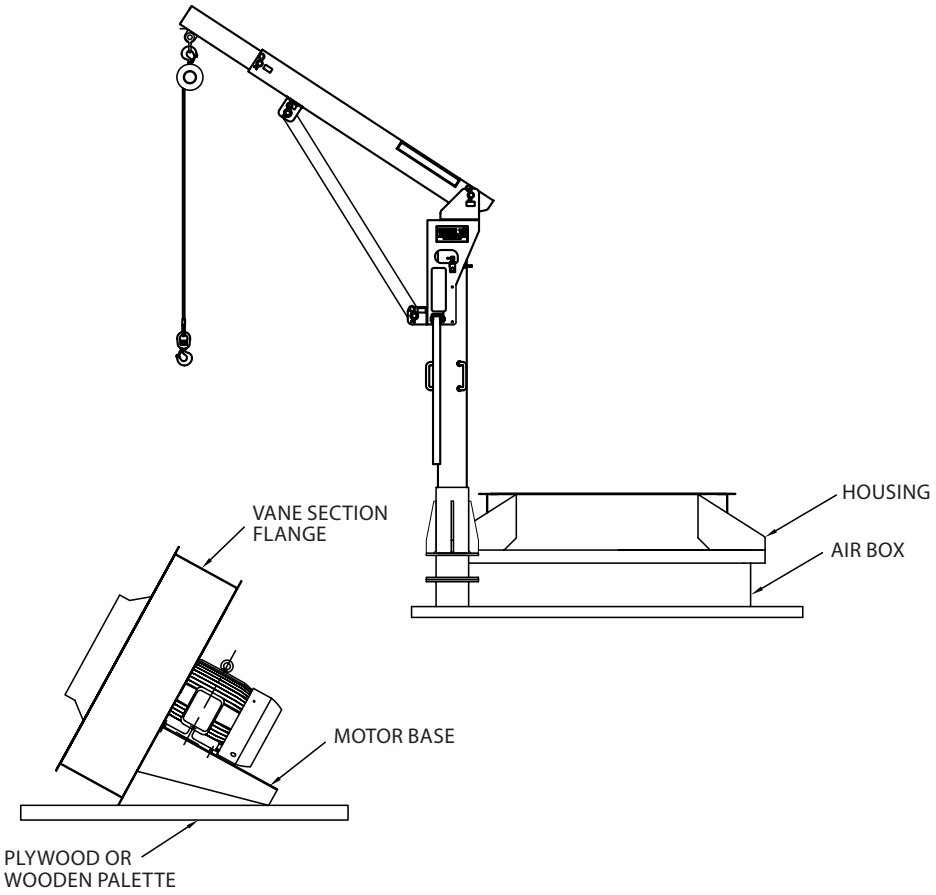
Lift Assembly High Enough to Clear Fan Housing

J-6



Lift Assembly High Enough to Clear Fan Housing

J-7



PROCEDURE FOR TRI-STACK WITH ACOUSTICAL NOZZLE MOTORCHANGE OUT USING BOLT-ON JIB CRANE

1) Disconnect all electric to the fan disconnect switch using standard lock out procedures.
2) Unbolt the disconnect switch from the vane section bracket and place it by the motor to allow for the nozzle to be removed without any obstruction.
3) Bolt jib post to the jib socket with 1/2-13x2" bolts. If this has a post instead of a plate, slide the post through the socket.
4) Remove the windband from the acoustical silencer nozzle.
5) Unbolt the housing from the vortex breaker, leaving the vane section, motor base and acoustical silencer nozzle bolted together.
6) Attach the lifting chains to the two lugs near the motor and the hole at the top of the motor base as the three picking points. ** DO NOT LIFT USING THE BRACKETS AT THE TOP OF THE ACOUSTICAL SILENCER NOZZLE**
7) The fan must be raised level, be sure that the chain at the hole in the motor base is short enough to provide a level and even lift.
8) Slowly raise the fan with the hoist. The fan only needs to be raised 2". Rotate the fan away from the plenum and lower it.
9) Lower the fan onto plywood or a wooden pallet.
10) When the fan has been fully lowered, unbolt the acoustical silencer nozzle from the vane section. Rotate the nozzle in place to break any adhesion to the vane section caused by caulk or gasket. Ensure all bolts have been removed from the nozzle to vane section.
11) Use the brackets on the acoustical silencer nozzle to lift the nozzle from the vane and fan section
12) Lower acoustical silencer nozzle onto plywood or a wooden pallet.
13) Unbolt the housing flange to the vane section.
14) Raise the fan once more, pull the housing away and lower the fan back down. The fan is now resting on the inlet of the fan blade.
15) Remove the chain from the lifting point on the motor base leaving two chains attached to the two lugs by the motor.
16) The fan needs to be tipped backwards and lean against the motor base. To do this, slowly lift the fan from the two lifting points. This can cause the fan to swing, be careful that the fan is clear of any objects.
17) Lower the fan and rest it on the motor base and the vane section top flange. The bottom of the motor should be facing the roof.
18) The blade is secured to the motor shaft by a single bolt keeper plate. Remove this bolt and pull the blade. (The jib can be used to assist this step if needed.)
19) Secure the jib hook to the eye-bolt of the motor. Slowly rotate the jib crank just until the cable is tight.
20) Unbolt and remove the motor.
21) Bolt the new motor in place but do not fully tighten the bolts for the motor feet.
22) Use feeler gages to check the clearance between the motor base and the motor feet. Use paper shims to maintain a 0.002-0.005 tolerance. After shimming the motor base, tighten the bolts for the motor feet.
23) Line up the keyway in the shaft with the keyway in the blade hub and install the blade. Be positive that the shaft key is in place prior to bolting on the keeper plate.
24) Pick up the fan again using the two lifting points. Now rotate the fan and lower it so it rests on the inlet of the fan blade again.
25) Place new gasket on the vane section flange, vortex breaker flange and housing flange.
26) Now lift the fan again from the three pick points and install the housing. Be sure that the inlet bell is centered in the blade inlet and that the blade can spin freely before tightening the bolts on the flange. (The alignment of the bell and blade can be done with the use of drift pins in the flange holes.)
27) Any wiring that needs to be done to either the motor or the disconnect switch should be done now before the nozzle is in place.
28) Place new gasket on the acoustical nozzle, vane section flange. Lift the acoustical nozzle from the two lifting eyes and replace it onto the vane/fan housing.
29) Re-attach the lifting chains to the two lugs near the motor and the hole at the top of the motor base as the three picking points. ** DO NOT LIFT USING THE BRACKETS AT THE TOP OF THE ACOUSTICAL SILENCER NOZZLE **
30) Place the fan back onto the vortex breaker and bolt in place.
31) Bolt the windband and the disconnect switch back into place.
32) Using silicone caulk, seal all mating flanges.
33) Run wires to the fan disconnect switch and "bump" start the fan to check the blade rotation is correct.
34) Once the rotation is correct, and the fan is up to full speed, check amps, volts and the vibration readings. Consult the factory if the fan is not performing to specification.

STROBIC AIR "TRI-STACK" FIELD FAN START-UP REPORT

STROBIC JOB NUMBER _____

JOB NAME _____

ADDRESS _____

CITY _____ STATE _____ ZIP CODE _____

PURCHASER _____

ADDRESS _____

CITY _____ STATE _____ ZIP CODE _____

PURCHASE ORDER NUMBER _____

CONTACT _____ TELEPHONE _____

NAMEPLATE DATA	User Fan Tag No.				
	Strobic Fan Tag No.				
	Date Manufactured				
	Fan Model No.				
	Fan Serial No.				
	Design CFM				
	Design ESP " W.G.				
	Motor Manufacture				
	Motor HP				
	Motor Type				
	Motor RPM				
	Electrical Data				
	(V/Cy/PH)				
	Motor Rated FLA				
	Location Served				
MEASURED DATA When fan is in operation	Volts L1-L2				
	Volts L2-L3				
	Volts L1-L3				
	RLA L1				
	RLA L2				
	RLA L3				

COMMENTS _____

START UP
PERFORMED BY _____

SIGNATURE _____

DATE _____

STANDARD STATEMENT OF TRI-STACKTM 7-YEAR FAN WARRANTY AND LIMITATION OF LIABILITY

At Strobic Air, we want you to benefit from a well-crafted and highly reliable system that you can utilize for years to come. We use the latest engineering and technical know-how to design and build Strobic Fans and Control Systems. In fact, we are so proud of how we engineer and build our products that we offer one of the most comprehensive warranties around.

Strobic Air warrants that products shall be shipped free from defects in materials and workmanship, and will operate in accordance with our proposals, specifications and nameplate data under proper conditions, installation, rated load, environment and usage for a period of **84 months (7 years)** from date of shipment from Strobic Air's location.

Strobic Air (Seller) is limited to the repair or replacement of the original parts. Buyer shall bear the costs of access (including removal and replacement of systems, structures or other parts of Buyer's facility), de-installation, installation, decontamination and transportation of defective Products to Seller. New or refurbished parts will be furnished FOB factory to Buyer's location at Seller's designated shipping point, freight allowed to Buyer's Location or ExWorks for shipments outside the conterminous United States.

These warranties and remedies are conditioned upon (a) the proper storage, installation, operation and maintenance of products and conformance with the proper operation instruction manuals provided by Seller or its suppliers or subcontractors, (b) Buyer keeping proper records of operation and maintenance during the warranty period and providing Seller access to those records, and (c) modification or repair of products only as authorized by Seller in writing. Seller does not warrant products or any repaired or replacement parts against normal wear and tear or damage caused by misuse, accident or use against the advice of Seller. Any modification or repair of any products not authorized by Seller shall render the warranty null and void.

Electrical components, excluding motors, are warranted only to the extent warranted by the original manufacturer. To the extent that Seller is entitled to pass through a warranty of the original equipment manufacturer of the electrical goods sold, Seller will pass through such warranties to Buyer. Strobic Air makes every effort to utilize materials that resist rust, but the warranty on Metal and Stainless Steel Components DOES NOT COVER RUST, OXIDATION, FADING or other BLEMISHES unless it also results in a loss of structural integrity or a failure of these components.

Any repair, replacement or reperformance by Seller hereunder shall not extend the applicable warranty period. The Parties shall mutually agree on the specifications of any test to determine presence of a defect. Seller shall have no liability for defects that arise after the warranty period has expired.

Made and engineered in the USA

1-800-SAC-FANS
strobicair.com

STROBIC AIR TECHNOLOGIES
A CECO Environmental Company

1966 50 2016
FIFTY YEARS OF INNOVATION